# Iterables, Iterators and Collections

Computer Science 112
Boston University

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# Collections

Java provides the internal mechanism that allow application work with and process a collection of Objects!

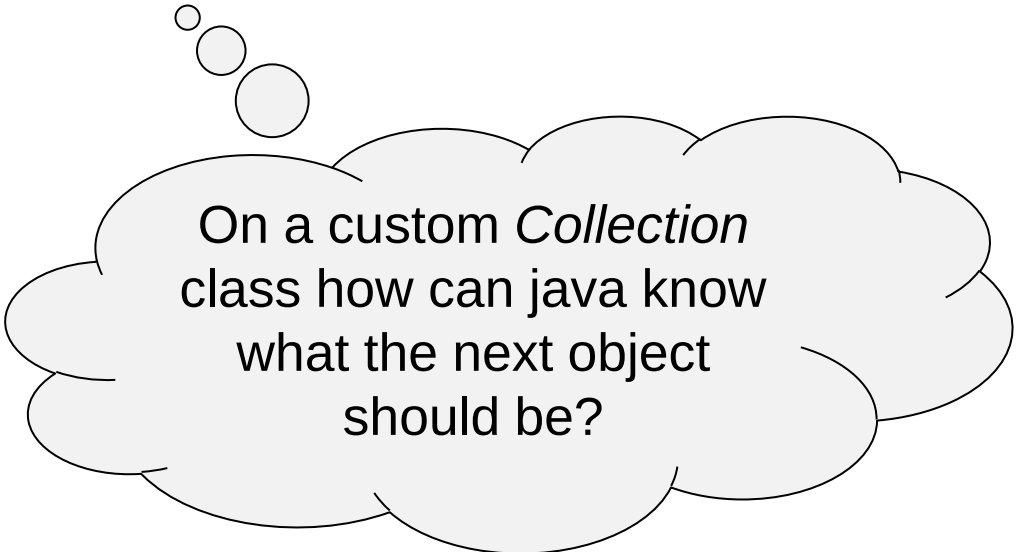
Specifically, Java allows you to work on any group of Objects as a single unit.

And all Collections share a common interface.

# Iteration Abstraction

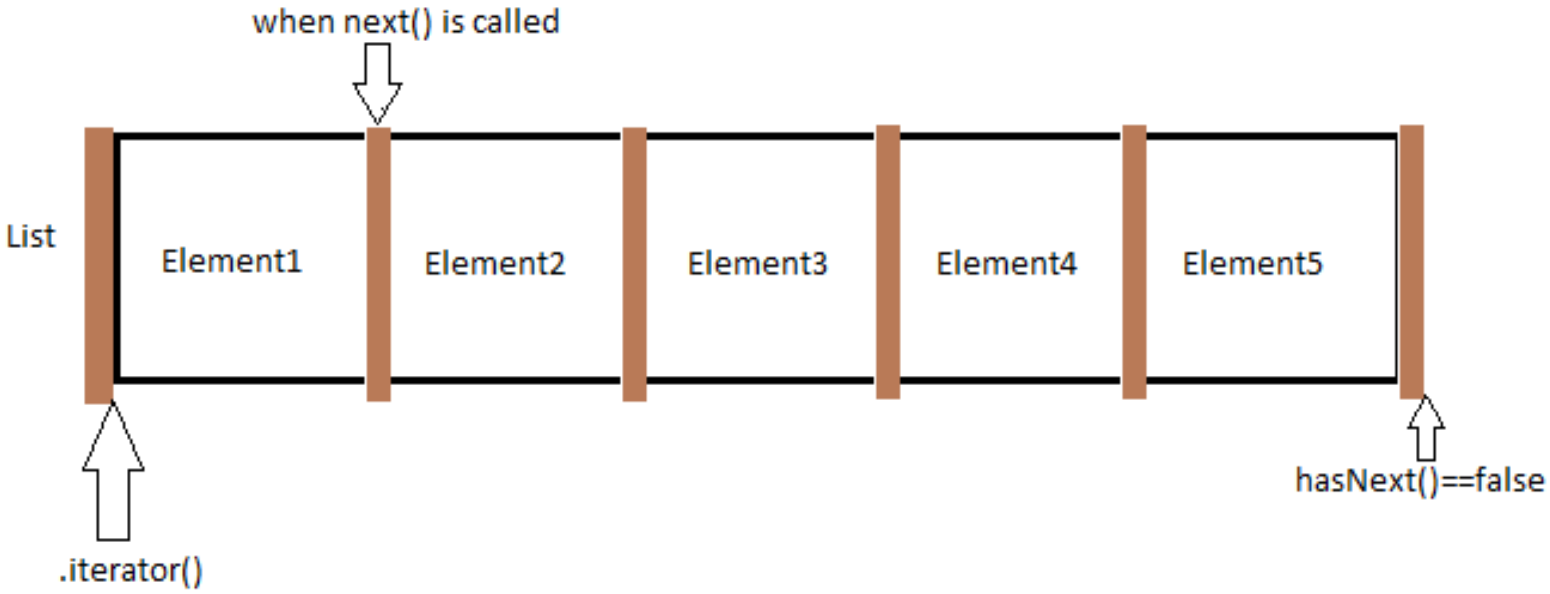
Iterators provide the ability to *iterate* over arbitrary types of data.

```
For all elements of the set
    Perform some action
```



# Iterators

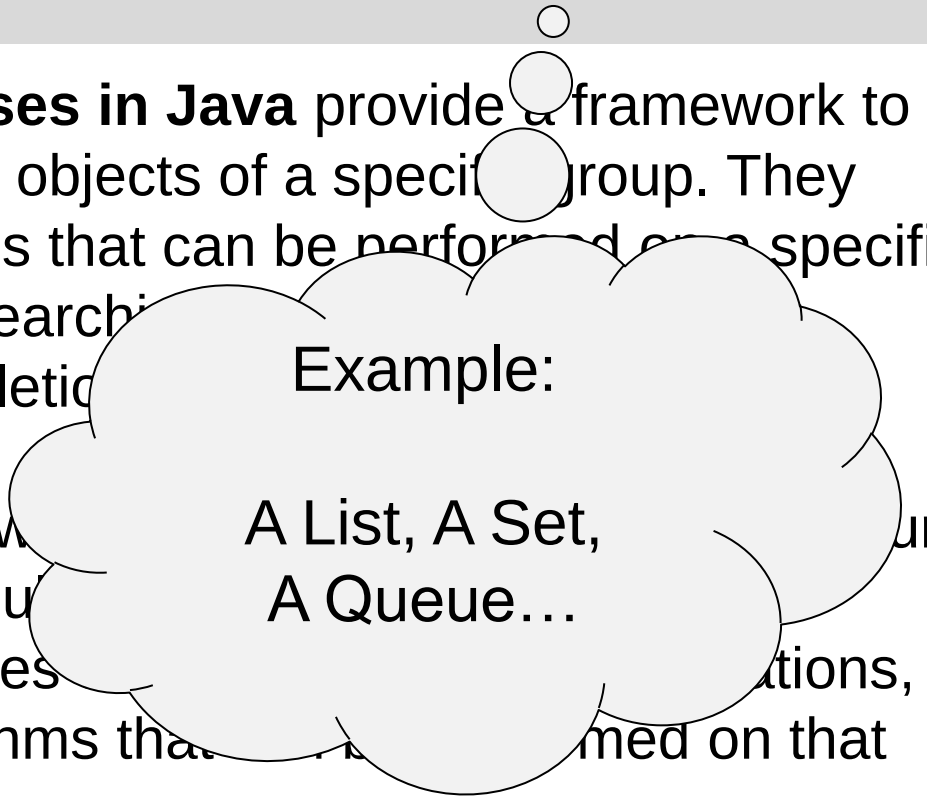
Iterators are used in ***Collection*** Classes in Java to retrieve (the elements of the Collection) one by one.



# Collection Classes

A Collection represents a single unit of objects, a group.

The **Collection classes in Java** provide a framework to *store* and *manipulate* objects of a specific group. They provide the operations that can be performed on a specific Collection, such as searching, manipulation, and deletion.




The Collection framework provides a unified architecture for storing and manipulating collections. It is comprised of Interfaces, implementations, along with the algorithms that can be performed on that collection.

Collection classes includes `ArrayList`, `LinkedList`, `PriorityQueue`, `HashSet`, to name a few.

# Collection Classes

A Collection represents a single unit of objects, a group.

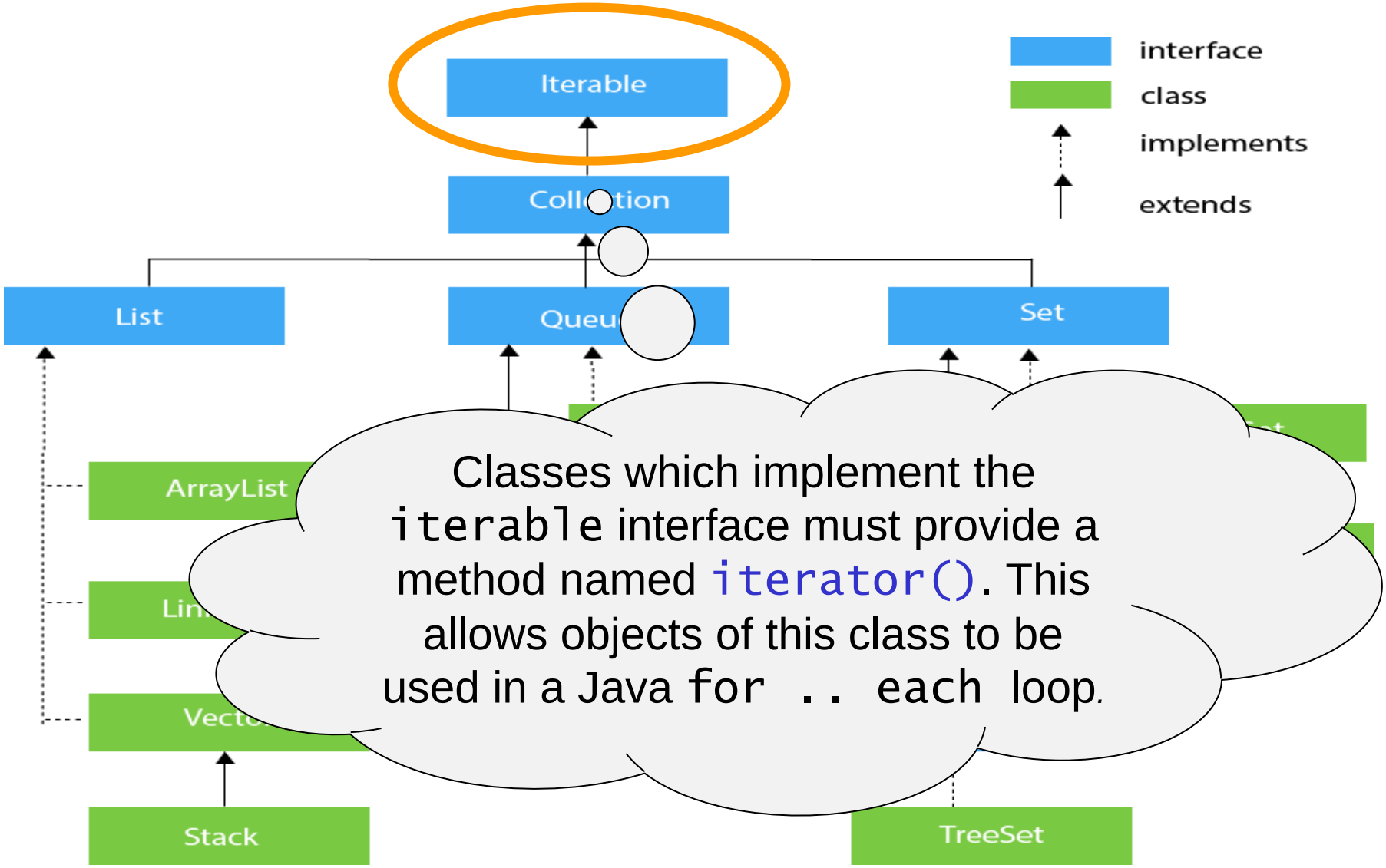
The **Collection classes in Java** provide a framework to *store* and *manipulate* objects of a specific group. They provide the operations that can be performed on a specific Collection, such as searching, sorting, insertion, manipulation, and deletion.

The Collection framework represents a unified architecture for storing and manipulating a group of objects. It is comprised of Interfaces and their class implementations, along with the algorithms that can be performed on that collection.

Collection classes includes `ArrayList`, `LinkedList`, `PriorityQueue`, `HashSet`, to name a few.

# Java Collection Classes

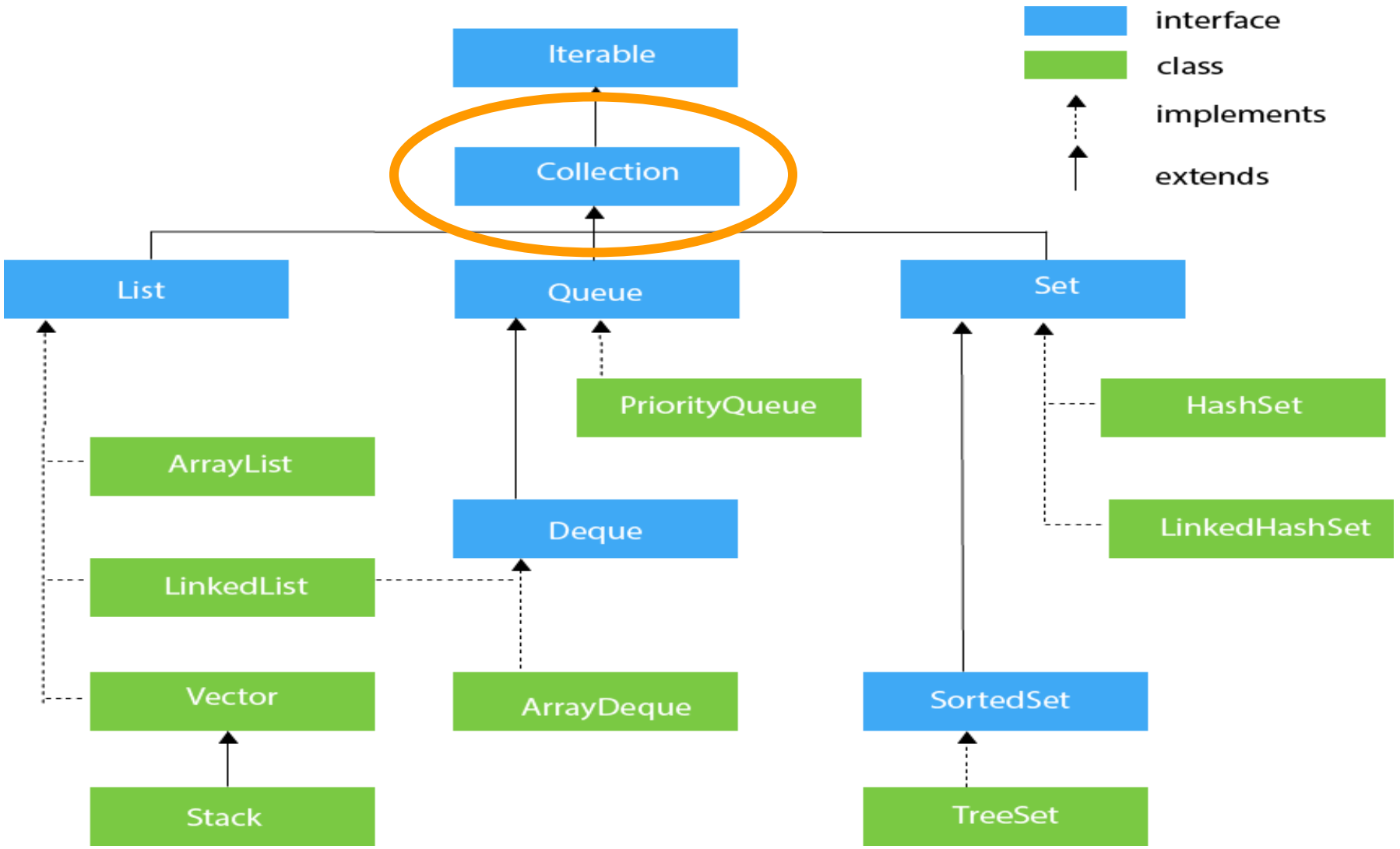
Java Tutorial Point




Classes which implement the `iterable` interface must provide a method named `iterator()`. This allows objects of this class to be used in a Java `for .. each` loop.

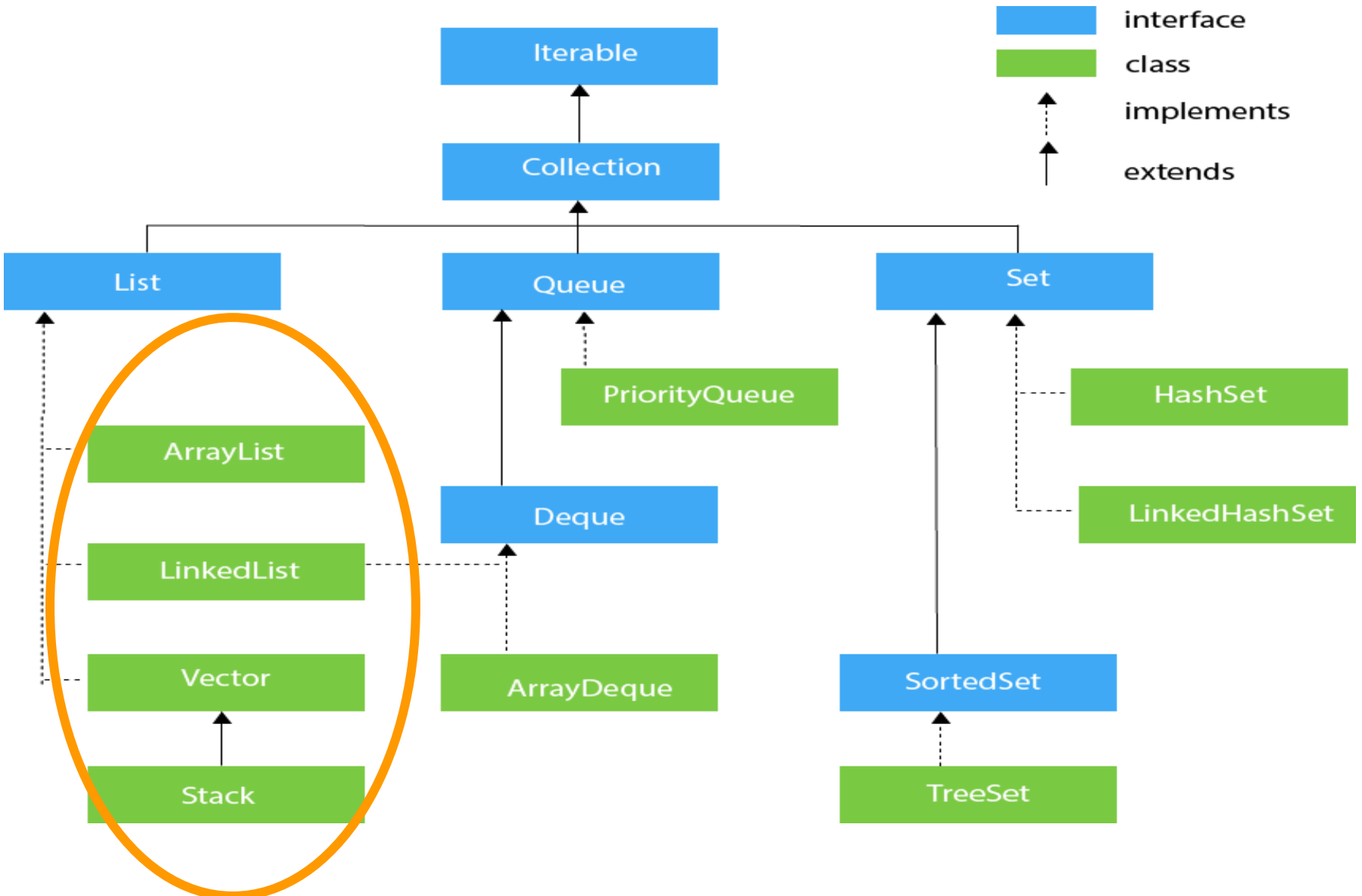
# Java Collection Classes

Java Tutorial Point



# Java Collection Classes

Java Tutorial Point



# Methods of the Collection Interface

Java Tutorial Point

The Collection interface is the interface which is implemented by all the classes in the collection framework.

| | |
|---|---|
| public boolean add(E e) | |
| public boolean addAll(Collection<? extends E> c) | |
| public boolean remove(Object element) | |
| public int size() | ction. |
| public void clear() | It removes the total number of elements from the collection. |
| public boolean contains(Object element) | It is used to search an element. |
| public boolean containsAll(Collection<?> c) | It is used to search the specified collection in the collection. |
| public Iterator iterator() | It returns an iterator. |
| public Object[] toArray() | It converts collection into array. |
| public boolean equals(Object element) | It matches two collections. |
| public int hashCode() | It returns the hash code number of the collection. |

# Methods of the Collection Interface

Java Tutorial Point

The Collection interface builds the foundation on which the collection framework depends. It declares the methods that every collection will have.

| | |
|---|---|
| **public boolean add(E e)** | |
| **public boolean addAll(Collection<? extends E> c)** | |
| **public boolean remove(Object element)** | |
| **public int size()** | |
| **public void clear()** | |
| **public boolean contains(Object element)** | It is used t n element. |
| **public boolean containsAll(Collection<?> c)** | It is used to search the specified collection in the collection. |
| **public Iterator iterator()** | It returns an iterator. |
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| **public boolean equals(Object element)** | It matches two collections. |
| **public int hashCode()** | It returns the hash code number of the collection. |

# Methods of the Collection Interface

Java Tutorial Point

| | |
|---|---|
| public boolean add(E e) | It is used to insert an element in this collection. |
| **public boolean addAll(Collection<? extends E> c)** | It is used to insert the specified collection elements in the invoking collection. |
| public boolean remove(Object element) | It is used to delete an element from the collection. |
| public int size() | It returns the total number of elements in the collection. |
| public void clear() | It removes the total number of elements from the collection. |
| public boolean contains(Object element) | It is used to search an element. |
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| public int hashCode() | It returns the hash code number of the collection. |

# Collection Classes

```
public class testClass {
    public static void main( String [] args ) {
        List<String> summer_fruits = new ArrayList<String>();




    }
}
```



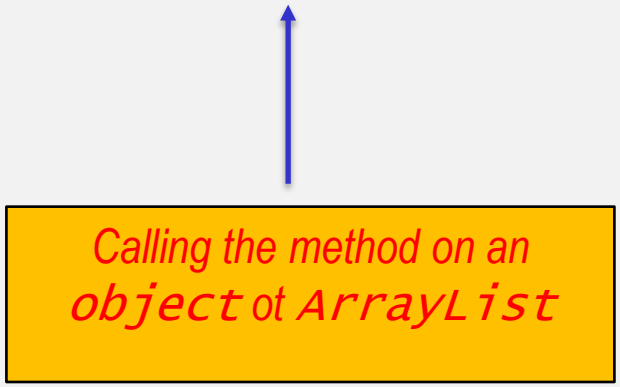
Interface

Class

# Collection Classes

```
public class testClass {
    public static void main( String [] args ) {
        List<String> summer_fruits = new ArrayList<String>();

        summer_fruits.add( “figs" );
        summer_fruits.add( "Mango" );
    }
}
```



# Collection Classes

```
public class testClass {
    public static void main( String [] args ) {
        List<String> summer_fruits = new ArrayList<String>();

        summer_fruits.add( “figs" );
        summer_fruits.add( "Mango" );

        List<String> fruits = new ArrayList<String>();

        fruits.addAll( summer_fruits );

    }
}
```

*Calling the method on an object ot ArrayList*

# Collection**S** Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> summer_fruits = new ArrayList<String>();

        summer_fruits.add( “figs" );
        summer_fruits.add( "Mango" );

        List<String> fruits = new ArrayList<String>();

        fruits.addAll( summer_fruits );

        Collections.addAll(fruits,"Apples","Oranges","Kiwi");

    }
}
```

*Calling a static method of the CollectionS class and passing an* ***object*** *ot* ***ArrayList***

# Collection**S** Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> summer_fruits = new ArrayList<String>();

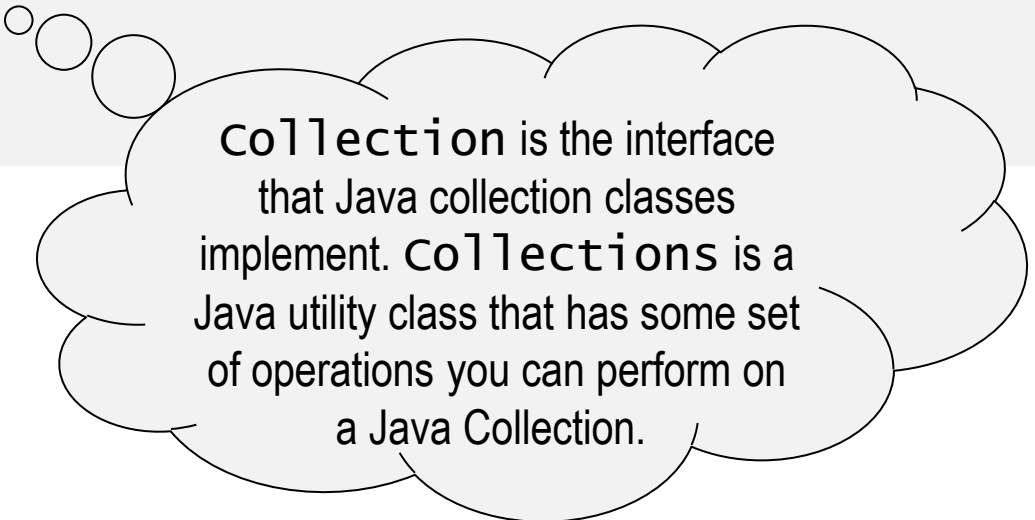
        summer_fruits.add( “figs" );
        summer_fruits.add( "Mango" );

        List<String> fruits = new ArrayList<String>();

        fruits.addAll( summer_fruits );

        Collections.addAll(fruits,"Apples","Oranges","Kiwi");

    }
}
```



# CollectionS Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

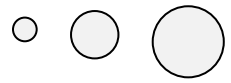
        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );




    }
}
```

```
Banana
Mango
Apples
Oranges
Kiwi
```



Note that the elements are output based on the order they were added to the Collection.

# CollectionS Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )       // element-based loop
            System.out.println( s );

        // What if we wanted to see the collection is some sorted order?


    }
}
```

# CollectionS Class

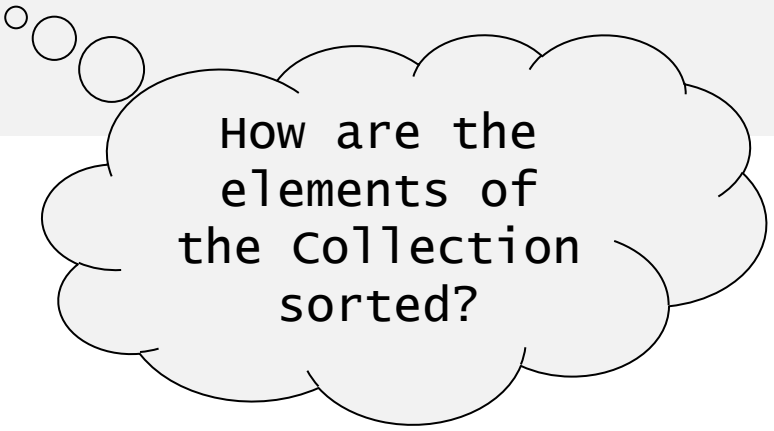
```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )       // element-based loop
            System.out.println( s );

        Collections.sort( fruits ); // Reorder the collection


    }
}
```



# CollectionS Class

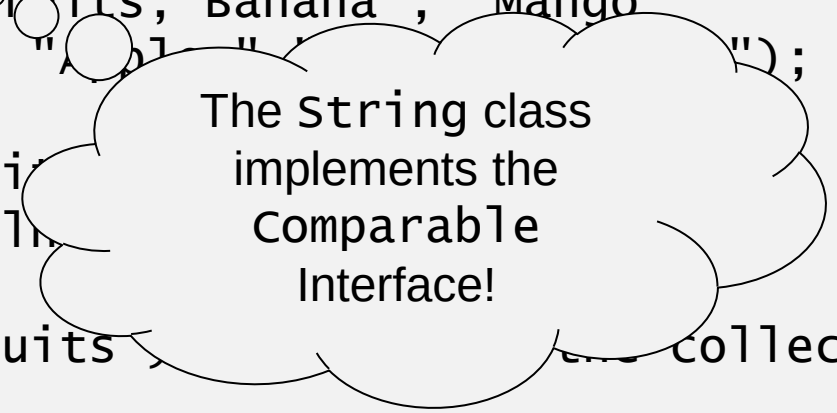
```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                                , "A                    ");

        for ( String s : frui                          loop
            System.out.println

        Collections.sort( fruits )                  collection


    }
}
```



# CollectionS Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );

        Collections.sort( fruits );
        for ( String s : fruits )        // element-based loop
            System.out.println( s );
    }
}
```

```
Apples
Banana
Kiwi
Mango
Oranges
```

Note that the elements are sorted based on their *natural order* in the `Collection`.

# CollectionS Class

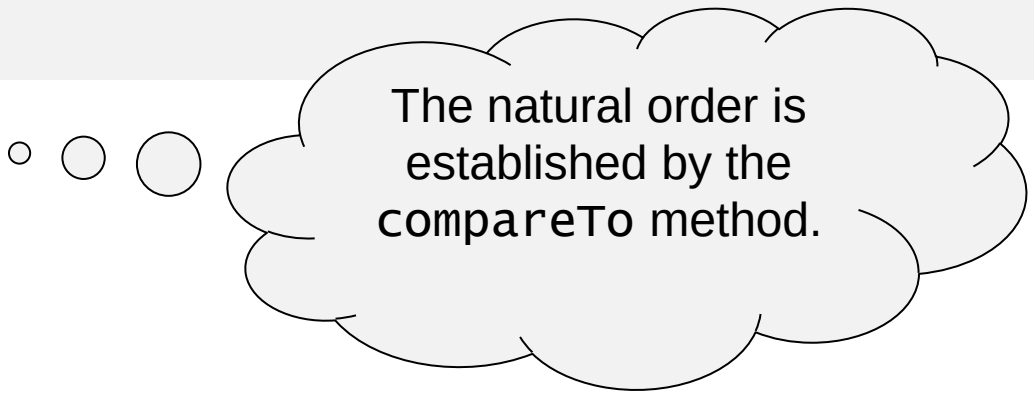
```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );

        Collections.sort( fruits );
        for ( String s : fruits )        // element-based loop
            System.out.println( s );
    }
}
```

```
Apples
Banana
Kiwi
Mango
Oranges
```



# CollectionS Class

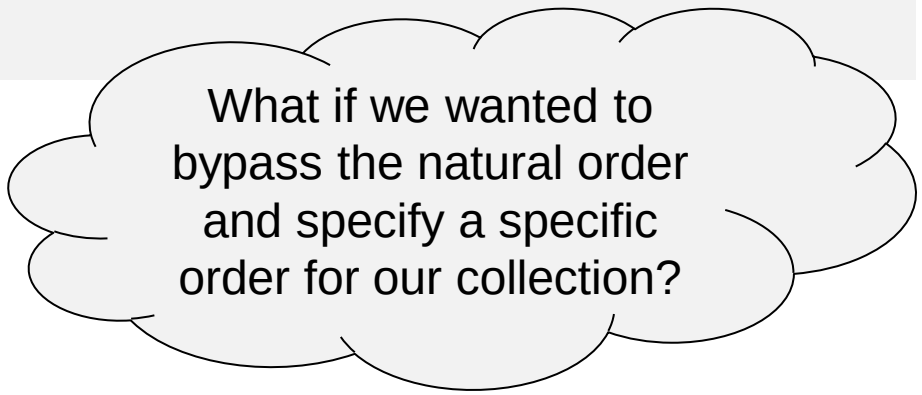
```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );

        Collections.sort( fruits );
        for ( String s : fruits )        // element-based loop
            System.out.println( s );
    }
}
```

```
Apples
Banana
Kiwi
Mango
Oranges
```



# **Comparator** Interface

```
public class lengthComparator implements Comparator<String>
{
    public int compare(String s1, String s2){

        return( s1.length() - s2.length() );


    }
} // class
```

# CollectionS Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );

        Collections.sort( fruits, new lengthComparator() );
        for ( String s : fruits )        // element-based loop
            System.out.println( s );
    }
}
```

```
Kiwi
Mango
Apples
Banana
Oranges
```

# CollectionS Class

```
public class testClass {
    public static voi
        List<String
        Collections
                                                    ","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );

        Collections.sort( fruits, new lengthComparator() );
        for ( String s : fruits )        // element-based loop
            System.out.println( s );
    }
}
```

Creating an instance of a class for the sole purpose calling a method on that instance.

```
Kiwi
Mango
Apples
Banana
Oranges
```

# CollectionS Class

```
public class testClass {
    public static voi
        List<String
        Collections
                                                    ");
                                                    go"
                                                    anges","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );

        Collections.sort( fruits, new lengthComparator() );
        for ( String s : fruits )        // element-based loop
            System.out.println( s );
    }
}
```

Strategy pattern!

```
Kiwi
Mango
Apples
Banana
Oranges
```

# **Comparator** Interface

```
public class lengthComparator implements Comparator<String>
{
    public int compare(String s1, String s2){

        return(s1.length() - s2.length());


    }
} // class
```

```
public class reverselengthComparator implements
Comparator<String>
{
    public int compare(String s1, String s2){

        return(s2.length() - s1.length();


    }
} // class
```

# CollectionS Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );

        Collections.sort(fruits, new reverselengthComparator());
        for ( String s : fruits )        // element-based loop
            System.out.println( s );
    }
}
```

```
Oranges
Apples
Banana
Mango
Kiwi
```

# CollectionS Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );

        Collections.sort( fruits );      // natural order
        for ( String s : fruits )        // element-based loop
            System.out.println( s );
    }
}
```

```
Apples
Banana
Kiwi
Mango
Oranges
```

# CollectionS Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );

        Collections.sort( fruits );
        fruits.forEach(System.out::println); // alternative

    }
}
```

```
Apples
Banana
Kiwi
Mango
Oranges
```

# CollectionS Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )        // element-based loop
            System.out.println( s );

        Collections.sort( fruits );
        fruits.forEach(System.out::println);
    }
}
```

```
Apples
Banana
Kiwi
Mango
Oranges
```

Internally the `forEach` method uses an element based `for-each` loop.

# CollectionS Class

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        Collections.addAll(fruits,”Banana”, “Mango”
                           , "Apples","Oranges","Kiwi");

        for ( String s : fruits )
            System.out.println( s );

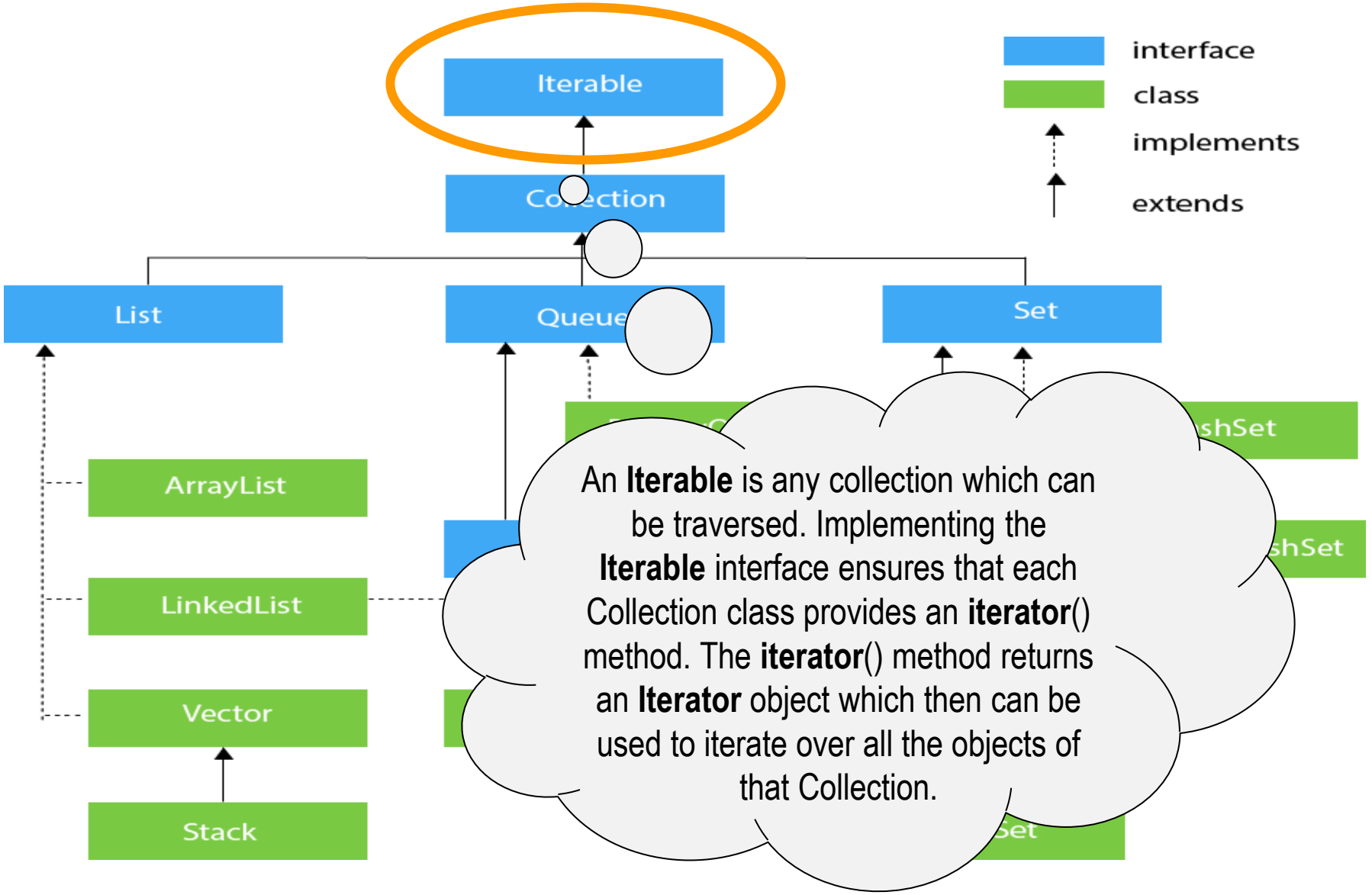
        Collections.sort( fruits );
        fruits.forEach(System.out::println);
    }
}
```

```
Apples
Banana
Kiwi
Mango
Oranges
```

How are the loops actually able to *iterate* over the objects in the `Collection`?

# Java Collection Classes

Java Tutorial Point



# Methods of the Collection Interface

Java Tutorial Point

| | |
|---|---|
| public boolean add(E e) | It is used to insert an element in this collection. |
| public boolean addAll(Collection<? extends E> c) | It is used to insert the specified collection elements in the invoking collection. |
| public boolean remove(Object element) | It is used to delete an element from the collection. |
| public int size() | It returns the total number of elements in the collection. |
| public void clear() | It removes the total number of elements from the collection. |
| public boolean contains(Object element) | It is used to search an element. |
| public boolean containsAll(Collection<?> c) | It is used to search the specified collection in the collection. |
| **public Iterator iterator()** | It returns an iterator. |
| public Object[] toArray() | It converts collection into array. |
| public boolean equals(Object element) | It matches two collections. |
| public int hashCode() | It returns the hash code number of the collection. |

# Collection Classes

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        fruits.add( "Banana" );
        fruits.add( "Mango" );
        Collections.addAll(fruits,"Apples","Oranges","Kiwi");

        for ( String s : fruits )
            System.out.println( s );

        Collections.sort( fruits );
        fruits.forEach(System.out::println);
    }
}
```

```
Apples
Banana
Kiwi
Mango
Oranges
```

Any class that implements the `Iterable` interface must provide an `iterator` method which creates an *iterator* object that is then used by the `forEach` loop.

# The Iterator Interface

*Provides the methods used to traverse the collection!*

# Using an Iterator

```
public class testClass {
    public static void main( String [] args ) {
        List<String> fruits = new ArrayList<String>();

        fruits.add( "Banana" );
        fruits.add( "Mango" );
        Collections.addAll(fruits,"Apples","Oranges","Kiwi");

        // Invoke the iterator method to create the iterator!
        Iterator itr = fruits.iterator();

        // check for availability of the next element
        while (itr.hasNext()) {
           // return the element at the current position and
           // move the cursor to next element
           System.out.println( (String) itr.next() );


    }
}
```

# The Iterator Interface

The Iterator interface provides the facility to create an iterator object which is used to *traverse* over the elements in the Collection, but in a forward direction only.

| Method | Description |
| --- | --- |
| public boolean hasNext() | It returns true if the iterator has more elements otherwise it returns false. |
| public Object next() | It returns the element and moves the cursor pointer to the next element. |
| public void remove() | It removes the last elements returned by the iterator. It is less used. |

# The Iterator Interface

The Iterator interface provides the facility to create an iterator object which is used to *traverse* over the elements in the Collection, but in a forward direction only.

| Method | Description |
| --- | --- |
| public boolean hasNext() | |
| public Object next() | |
| public void remove() | ed by the it d. |

Known as the Universal iterator because it is the core interface of all the Collection classes.

```
public class TestIterator
{
    public static void main(String[] args) {
        // Create an array list
        ArrayList al = new ArrayList();

        // Add the numbers 0 .. 9 to the list
        for (int i = 0; i < 10; i++)
            al.add(i);

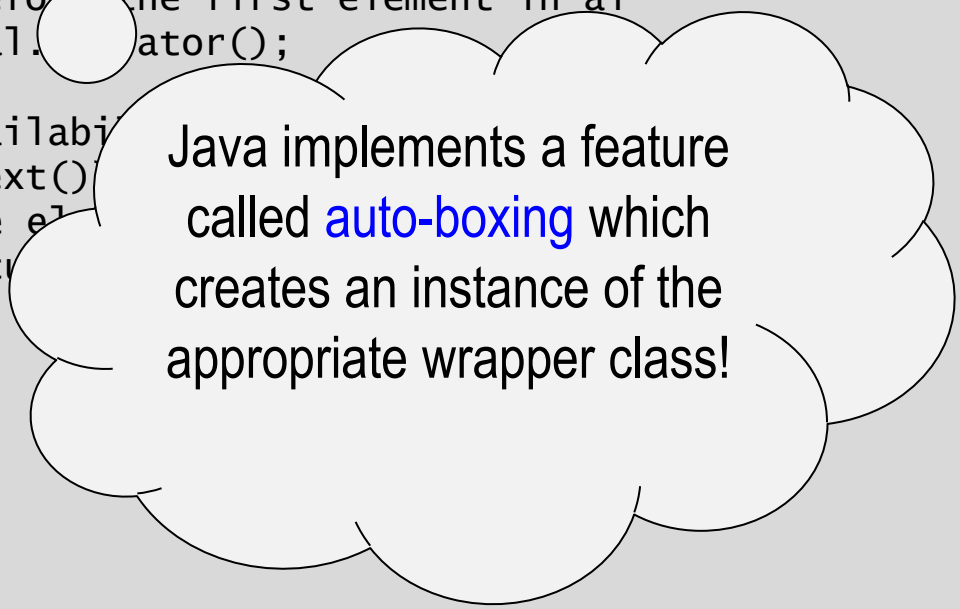
        // at beginnin itr(cursor) will point to
        // index just befo the first element in al
        Iterator itr = al. ator();

        // check for availabi
        while (itr.hasNext()
           // return the el
           // move the cu
           int i = (int)

        }

    } // main

} // class
```



```
public class TestIterator
{
    public static void main(String[] args) {
        // Create an array list
        ArrayList al = new ArrayList();

        // Add the numbers 0 .. 9 to the list
        for (int i = 0; i < 10; i++)
            al.add(i);

        // at beginning itr(cursor) will point to
        // index just before the first element in al
        Iterator itr = al.iterator();

        // check for availability of the next element
        while (itr.hasNext()) {
           // return the element at the current position and
           // move the cursor to next element
           int i = (int)itr.next();

           // Can even remove elements while iterating
           if (i % 2 != 0)
              itr.remove();
        }

    } // main

} // class
```

# ListIterator Interface **extends** Iterator

The List Iterator interface provides the facility of iterating over List style collection classes that provides bi-directional iteration.

```
ListIterator ltr = l.listIterator();
```

There are two additional methods that are provided by the ListIterator Interface:

| Method | Description |
|---|---|
| public boolean hasPrevious() | It returns true if the iterator has more elements while traversing backward otherwise it returns false. |
| public Object previous() | It returns the previous element in the iteration and moves the cursor pointer to the next previous element. |

# Iterator vs. ListIterator

## *summary*

- The basic difference between Iterator and ListIterator is that the Iterator can traverse elements in a collection only in forward direction. On the other hand, the ListIterator can traverse in both forward and backward directions.
- Using iterator you can not add any element to a collection. But, by using ListIterator you can add elements to a collection.
- Using Iterator, you can not remove an element in a collection where, as you can remove an element from a collection using ListIterator.
- Using Iterator you can traverse all collections like *Map*, *List*, *Set*. But, by ListIteror you can traverse List implemented objects only.

# Implementing the List Iterator Interface

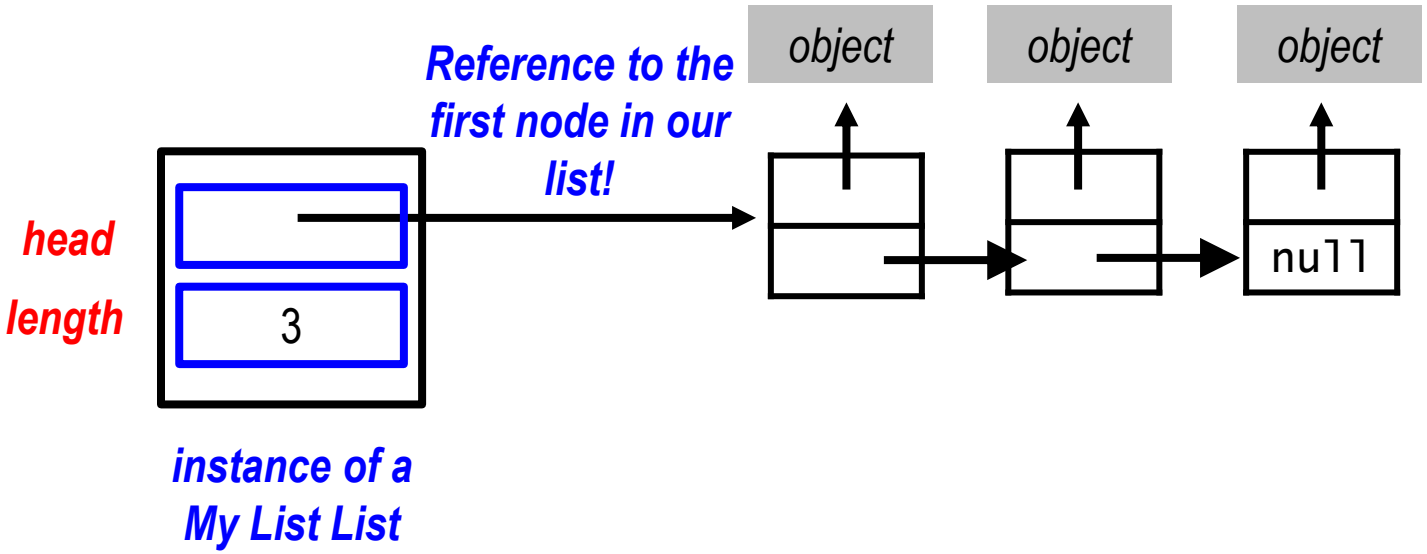
- Here again, the interface only includes the method headers:

```
public interface ListIterator { // in ListIterator.java
    boolean hasNext();
    Object next();
}
```

- We can then implement this interface for our own list:
  - Assume a class MyList that simulates a linked lists

# MyList Class

- Implementing the List interface with a **Linked List**



# A Linked List class

```
public class MyList implements List, Iterable {
    private class Node {
        private Object item;
        private Node next;

        private Node() {
            next = null;
        }
    }
    private Node head;
    private int length;

    public MyList() {
        head = null; length = 0;
    }
    public boolean add( Object item ) { ... };
    public Object remove( Object item ) { ... }; etc.

    public ListIterator iterator() {
        return new MyListIterator();
    }

}
```

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    private class Node {
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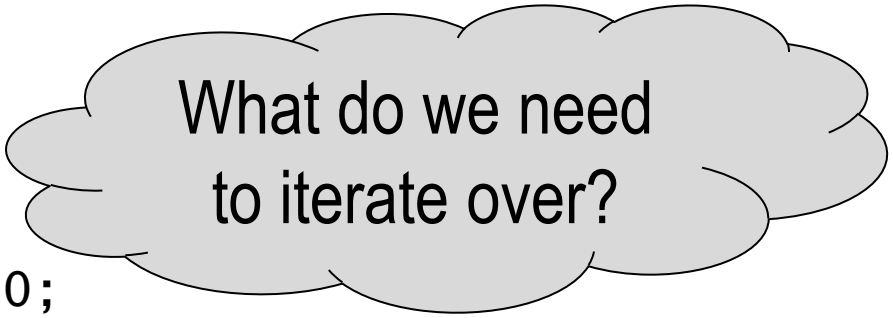

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```
public class MyList implements List, Iterable {
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        private Object item;
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        private Node() {
            next = null;
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    }
    private Node head;
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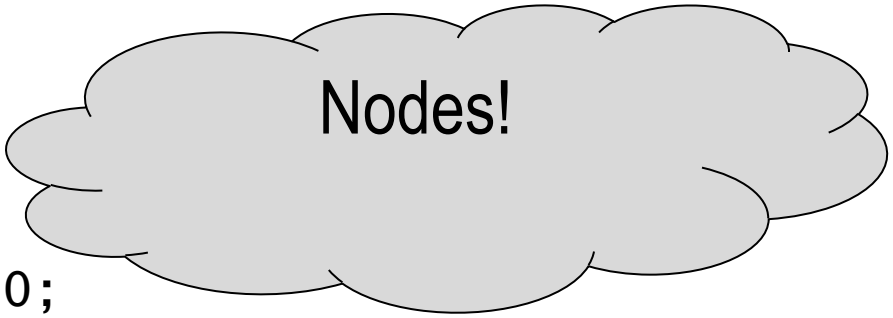


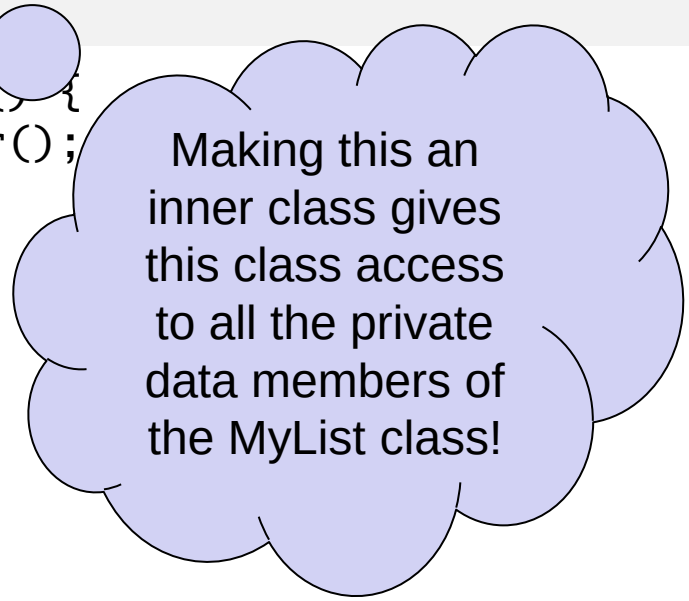
# An Inner Class for the Iterator

```
public class MyList … {
    private Node head;
    private int length;

    private class MyListIterator implements ListIterator {
        private Node nextNode;  // points to node with the next item

        public MyListIterator() {
            nextNode = head;
        }
        ...
    }

    public ListIterator iterator() {
        return new MyListIterator();
    }
    ...
```



# Full LLListIterator Implementation

```
private class MyListIterator implements ListIterator {
    private Node nextNode;     // points to node with the next item

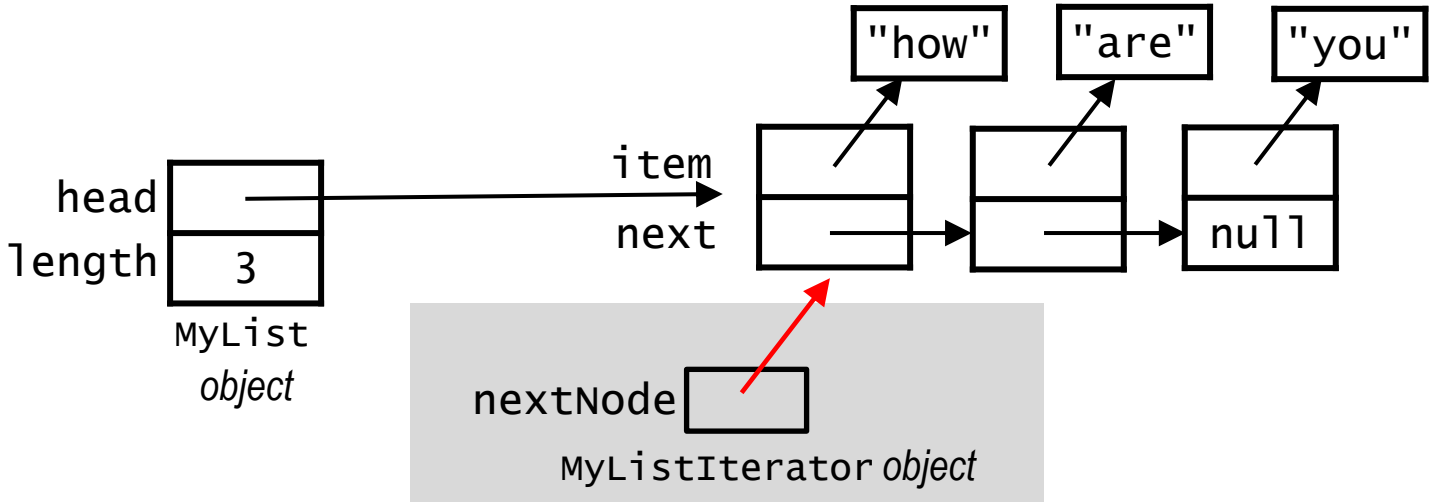
    public MyListIterator() {
        nextNode = head;
    }

    public boolean hasNext() {
        return (nextNode != null);
    }

    public Object next() {
        // throw an exception if nextNode is null

        Object item = nextNode.item;
        nextNode = nextNode.next;

        return item;
    }
}
```



# An Interface for List Iterators:

*summary*

Once the iterator interface has been implemented, we can create an instance of it and use it to externally traverse the list - regardless of the specific implementation of the List:

```
public class MyClass {

   public static int numOccur(List l, Object item) {

      int numOccur = 0;
      ListIterator iter = l.iterator();
      while ( iter.hasNext() ) {
         Object itemAt = iter.next();

         if (itemAt.equals(item)) {
            numOccur++;
         }
      }
      return numOccur;
   } ...
```

# Example: *MyList* list

```
public class MyClass {
    public static int numOccur(List l, Object item) {
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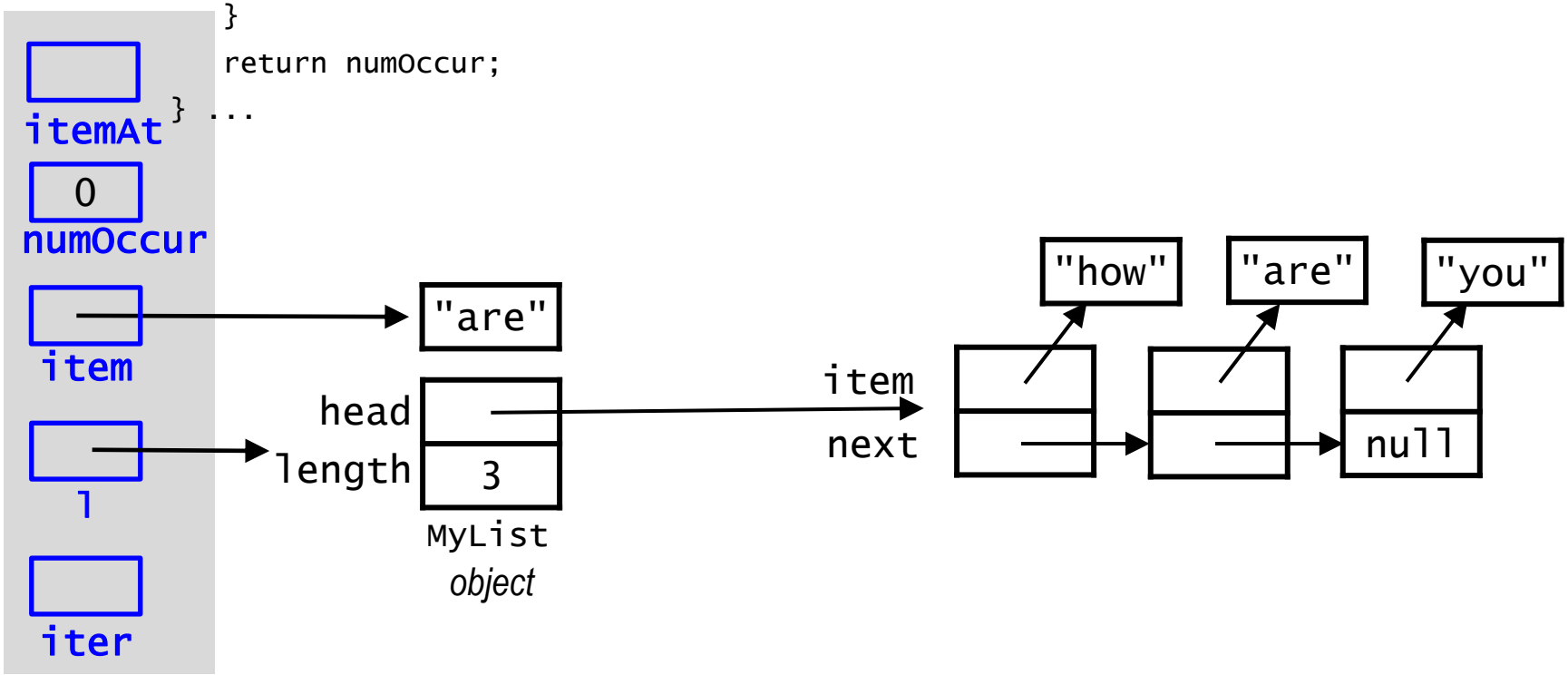

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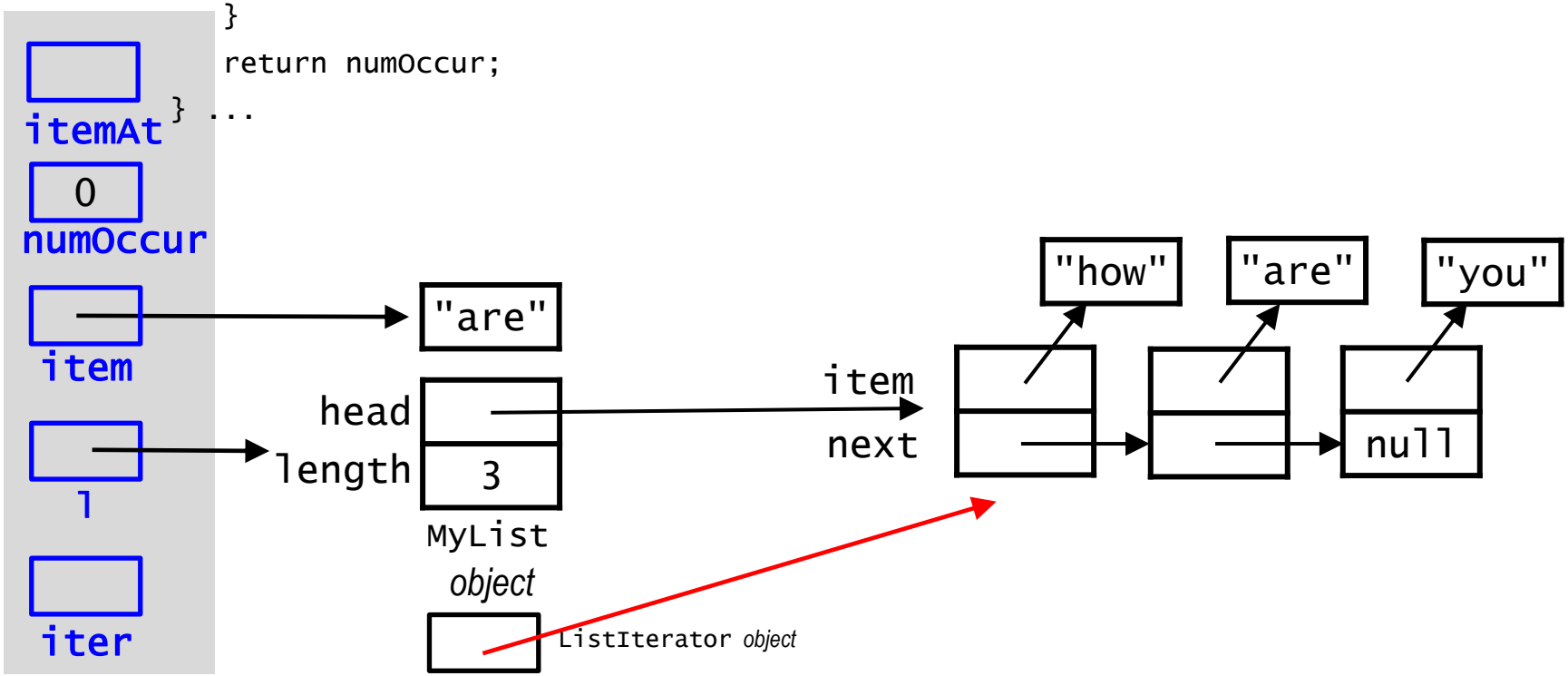

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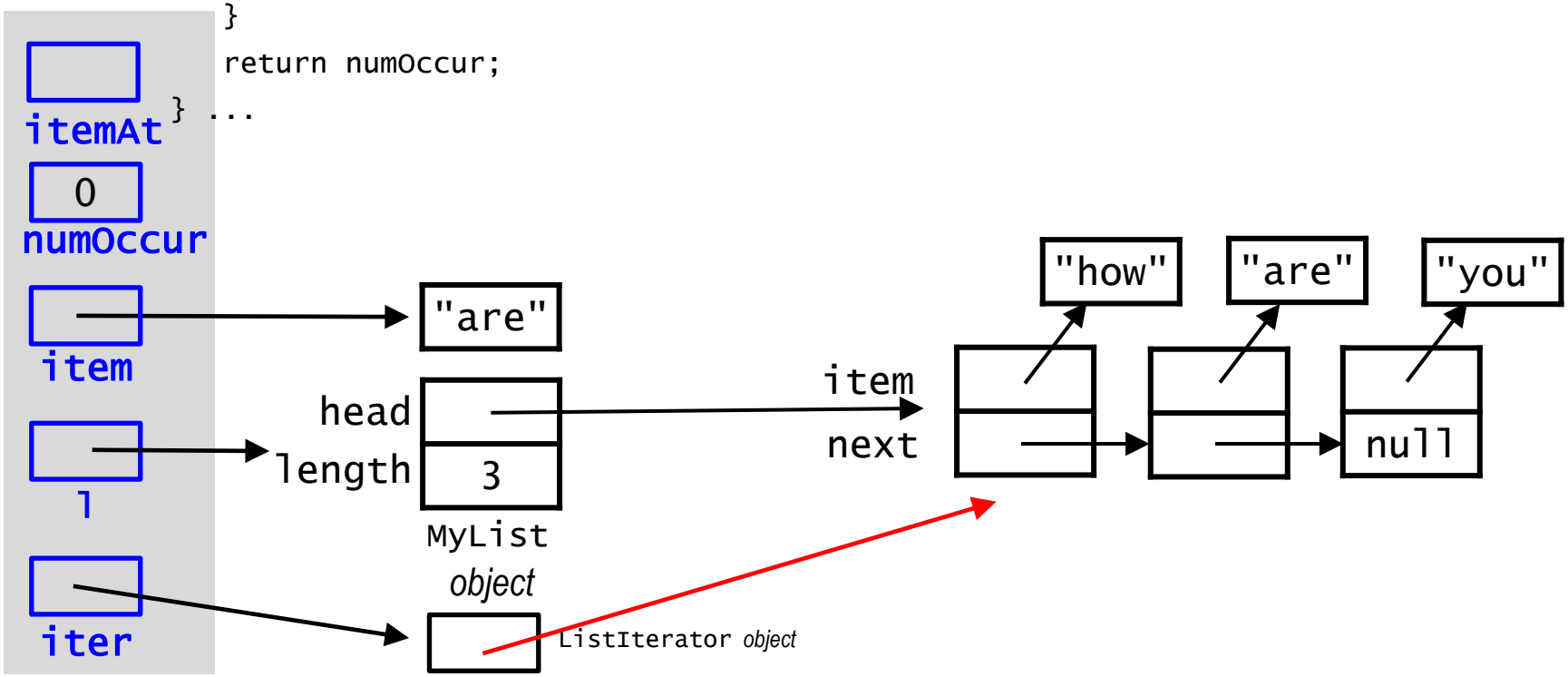

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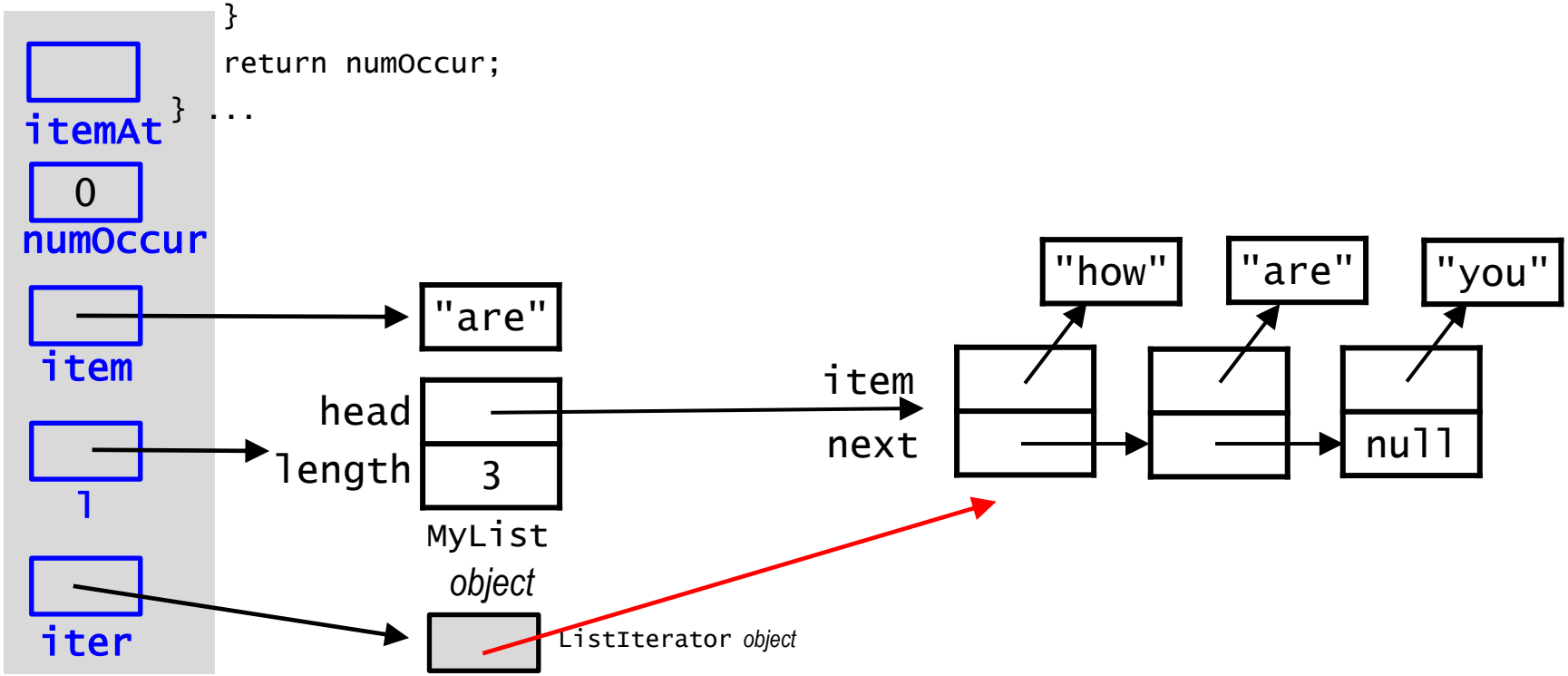

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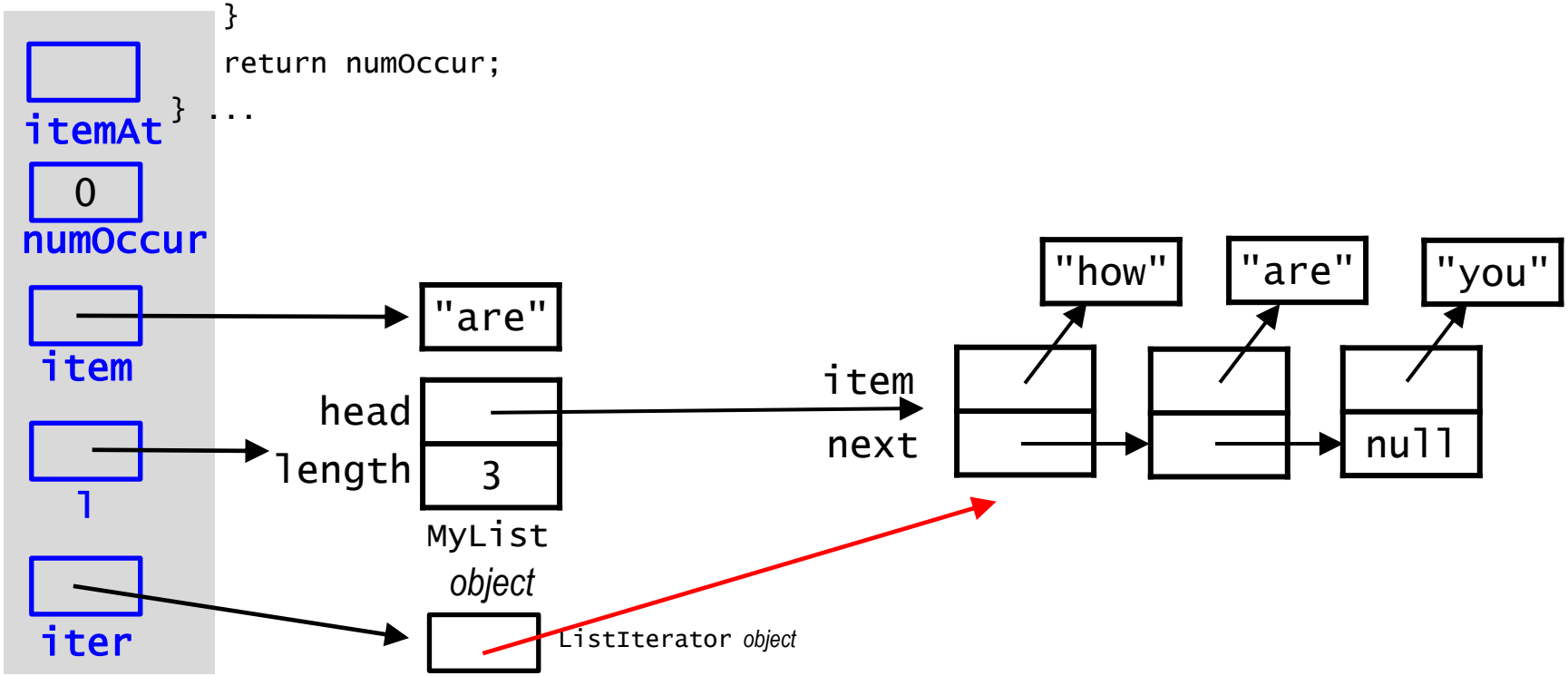

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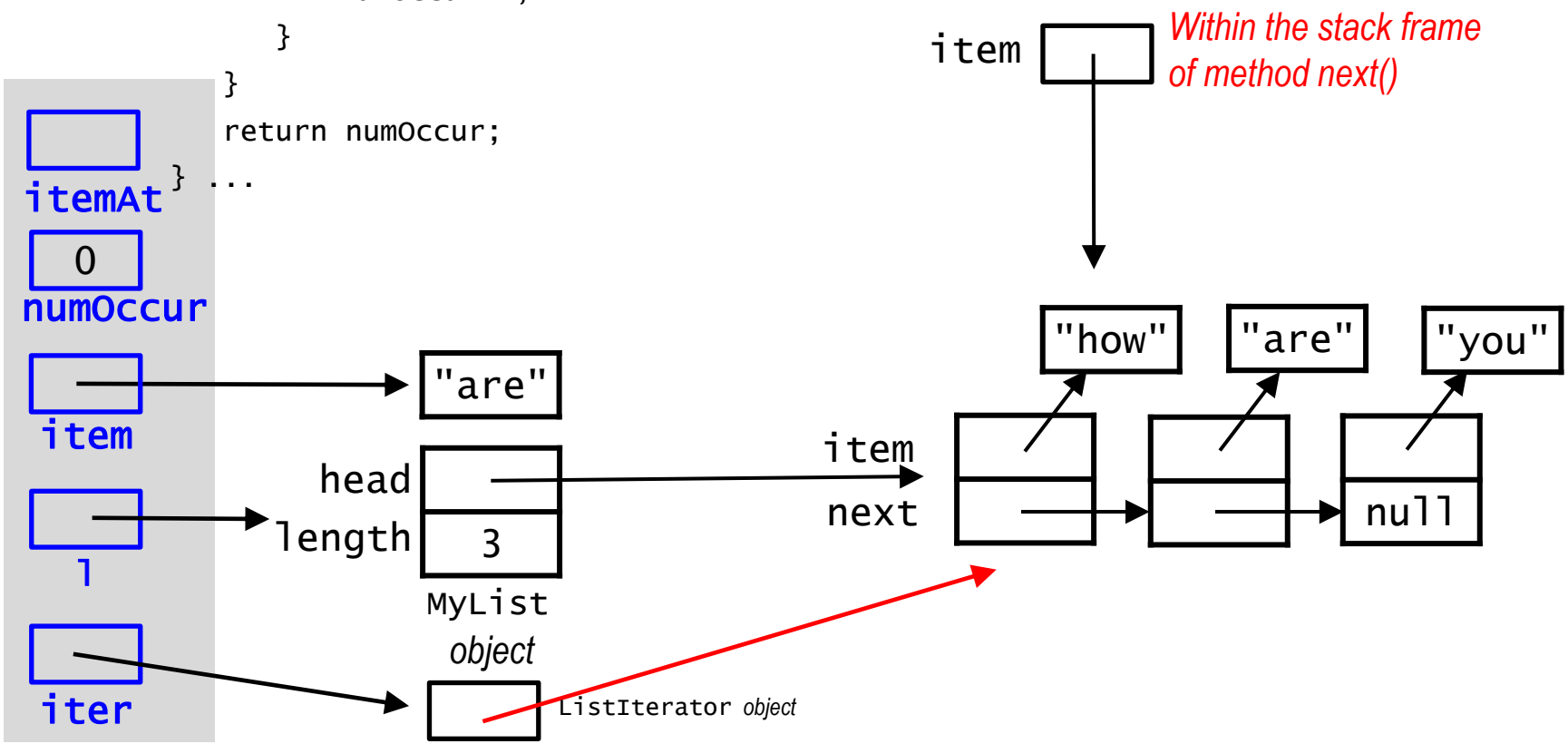

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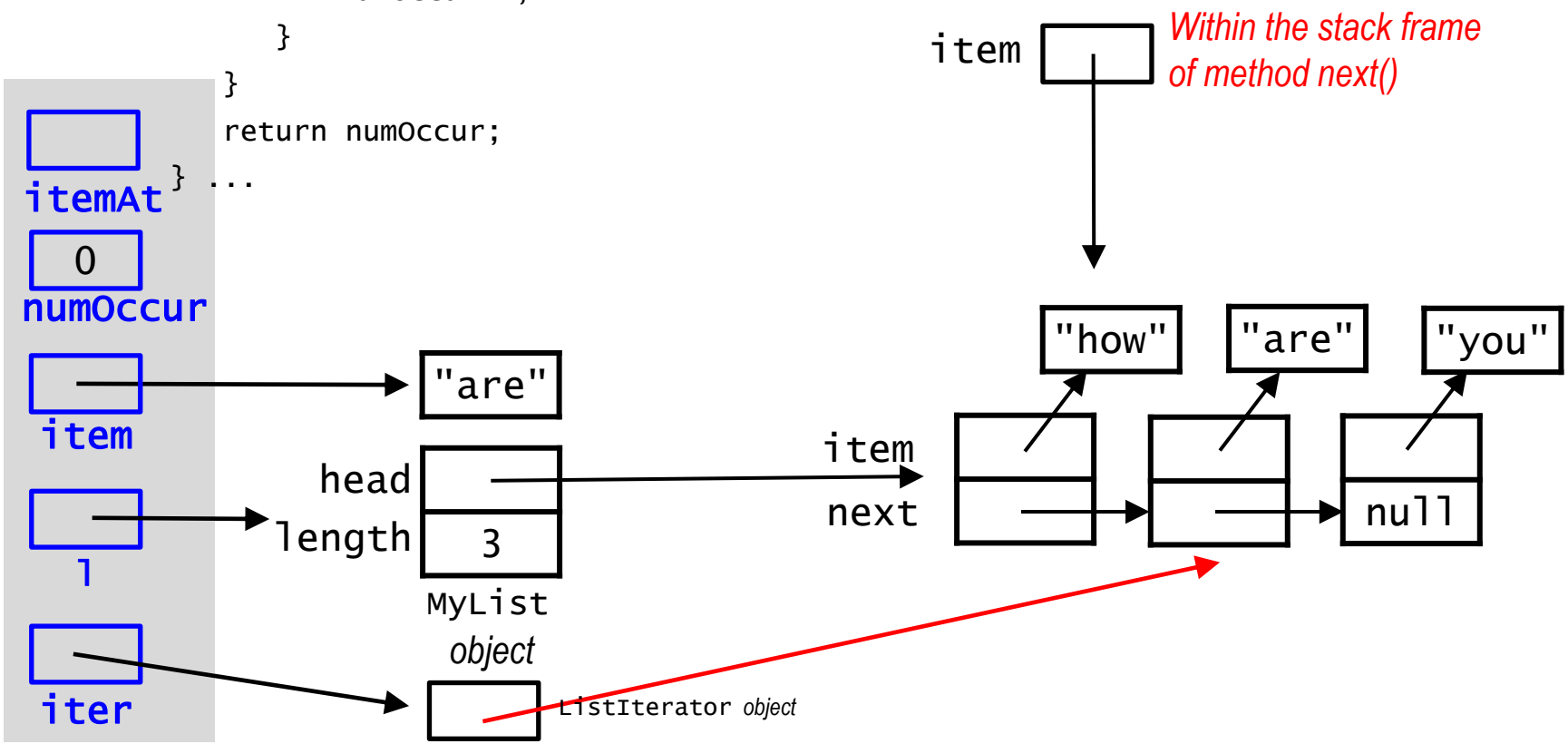

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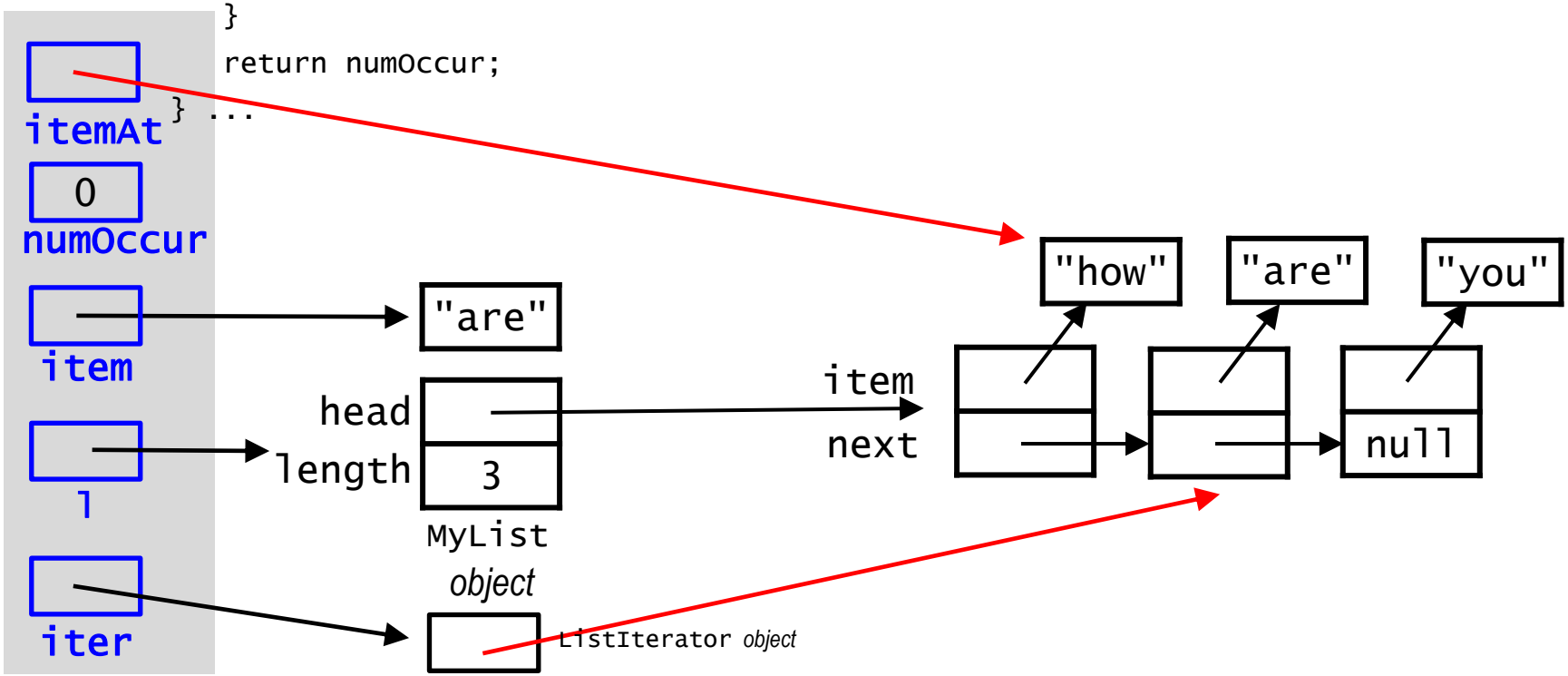

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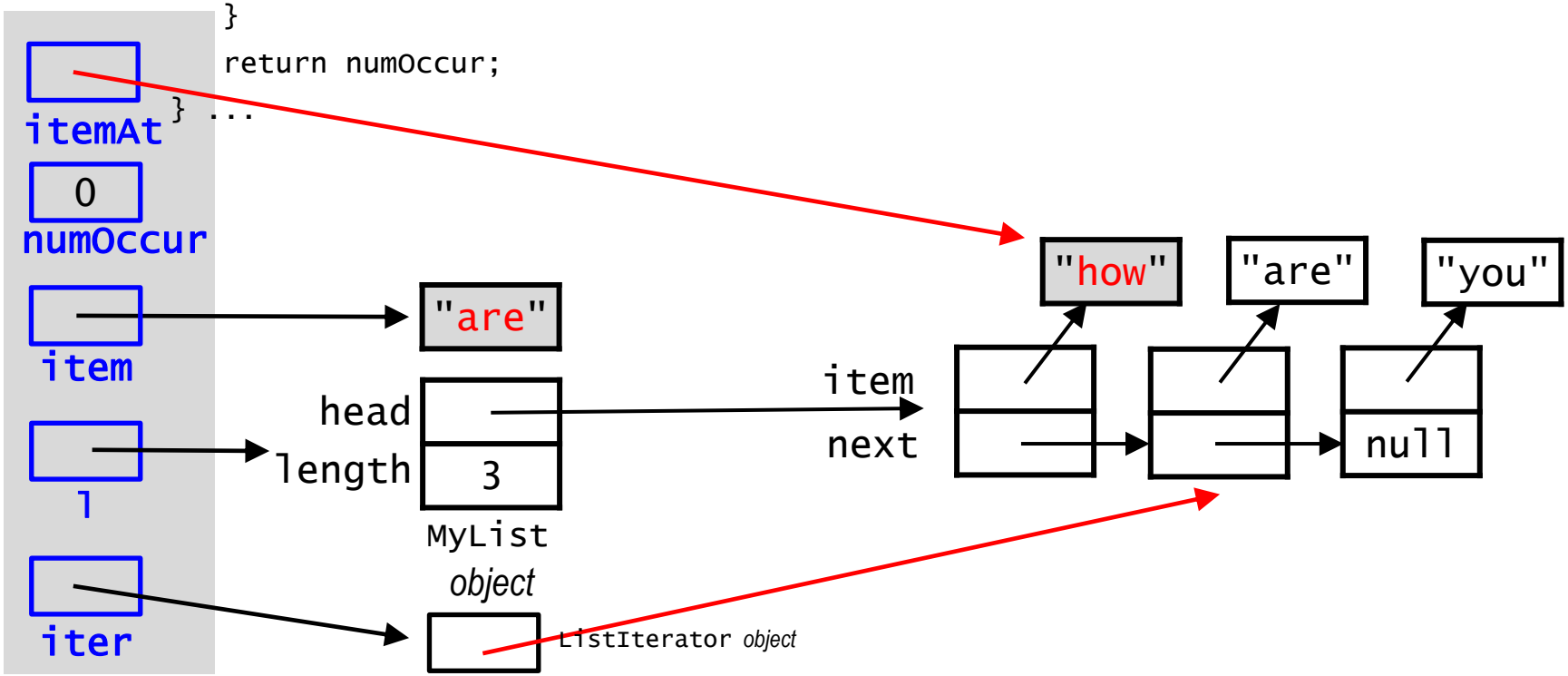

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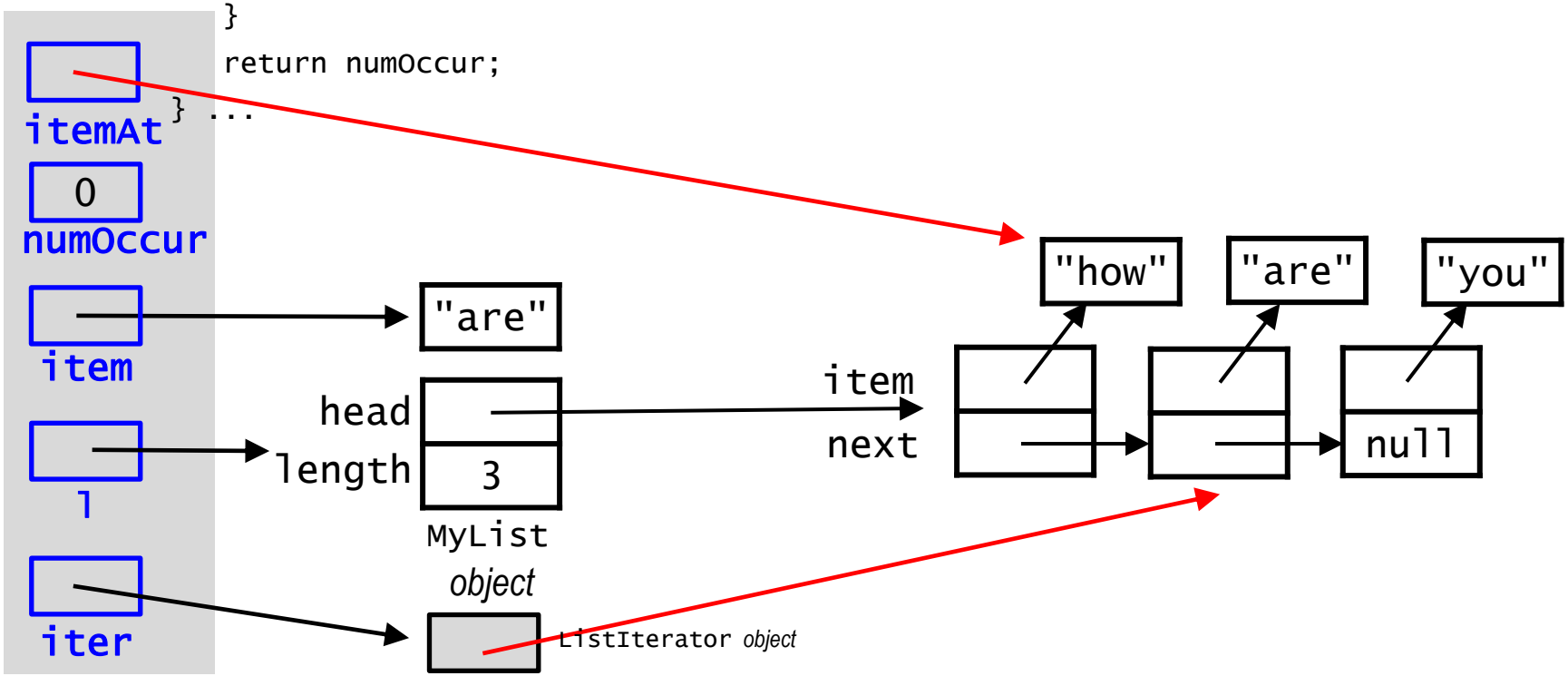

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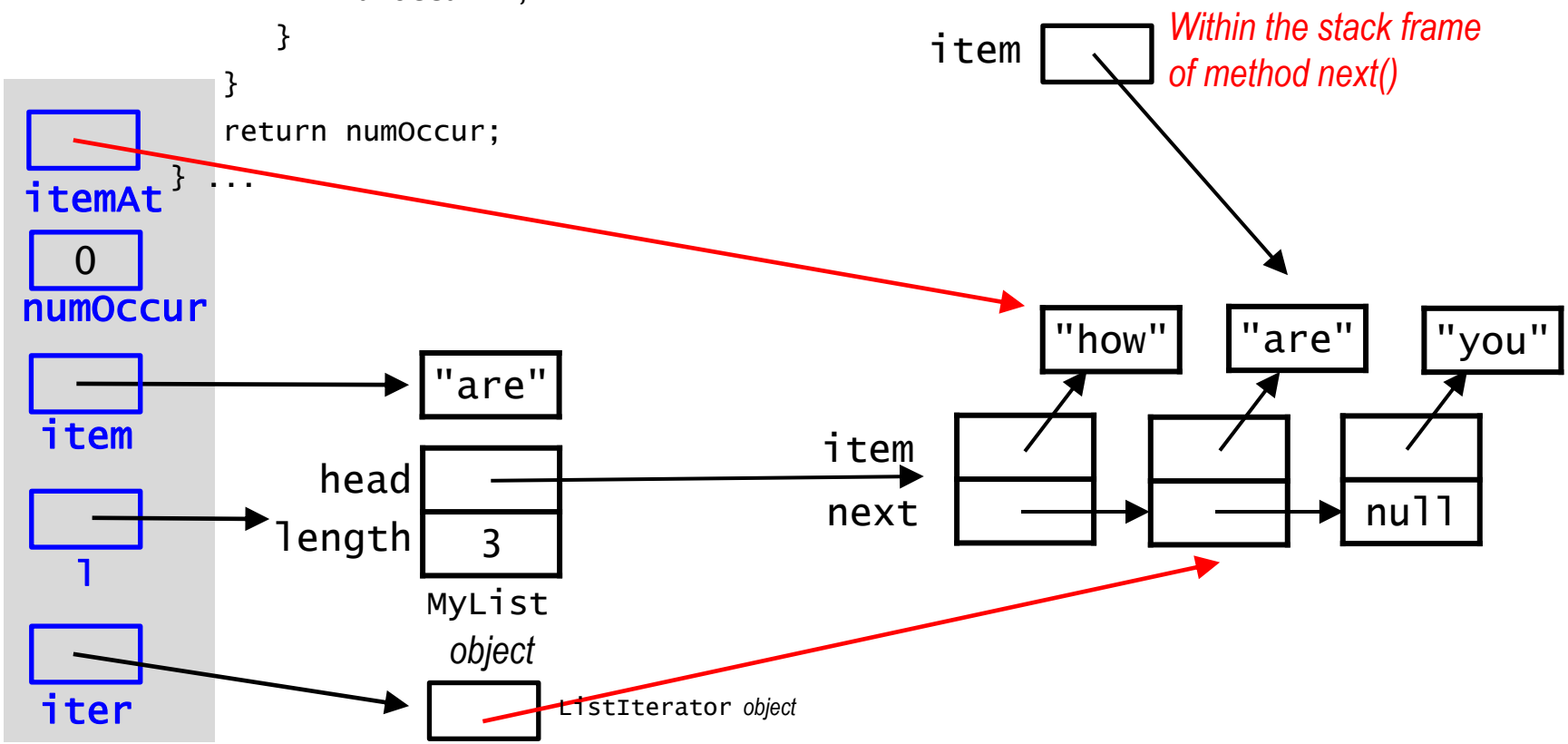

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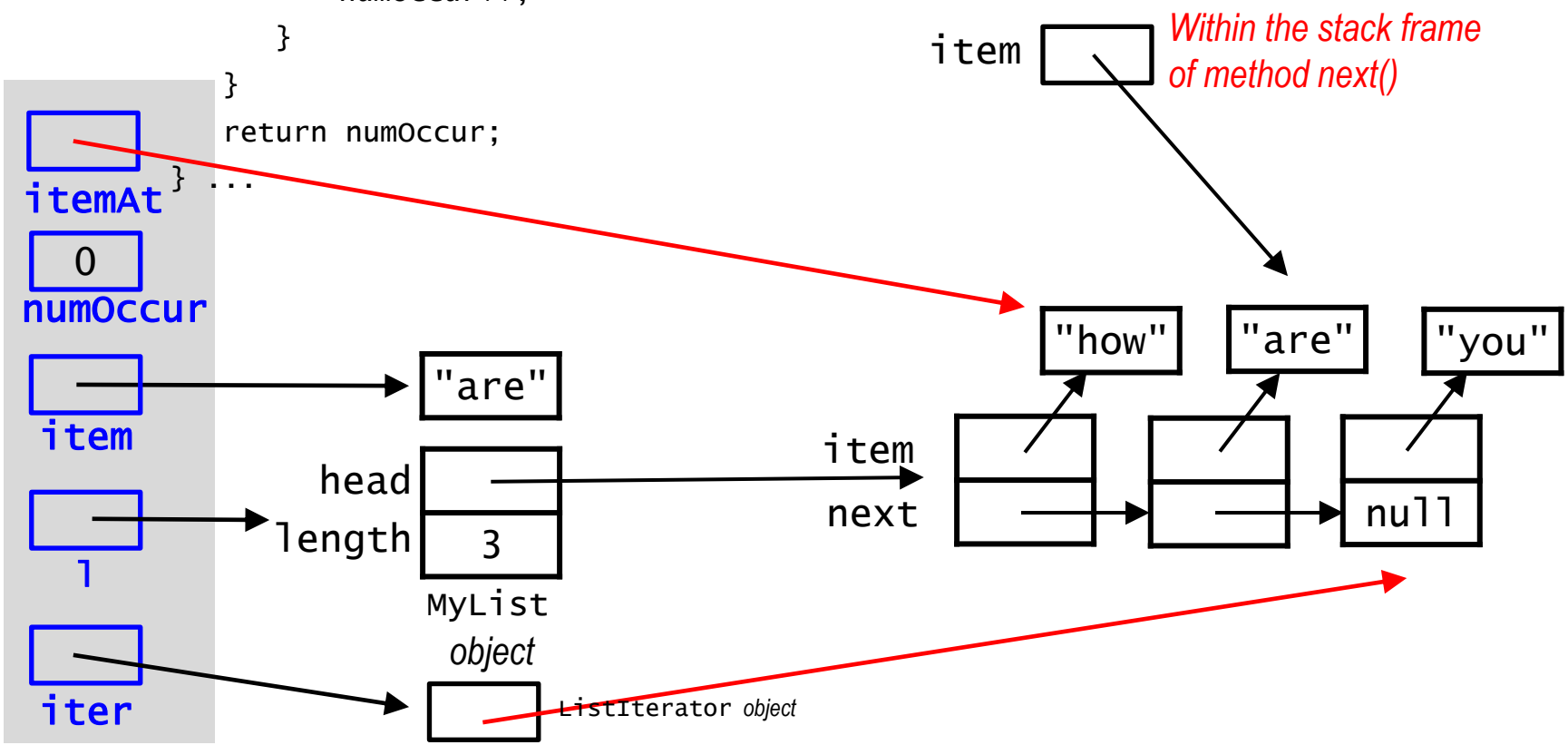

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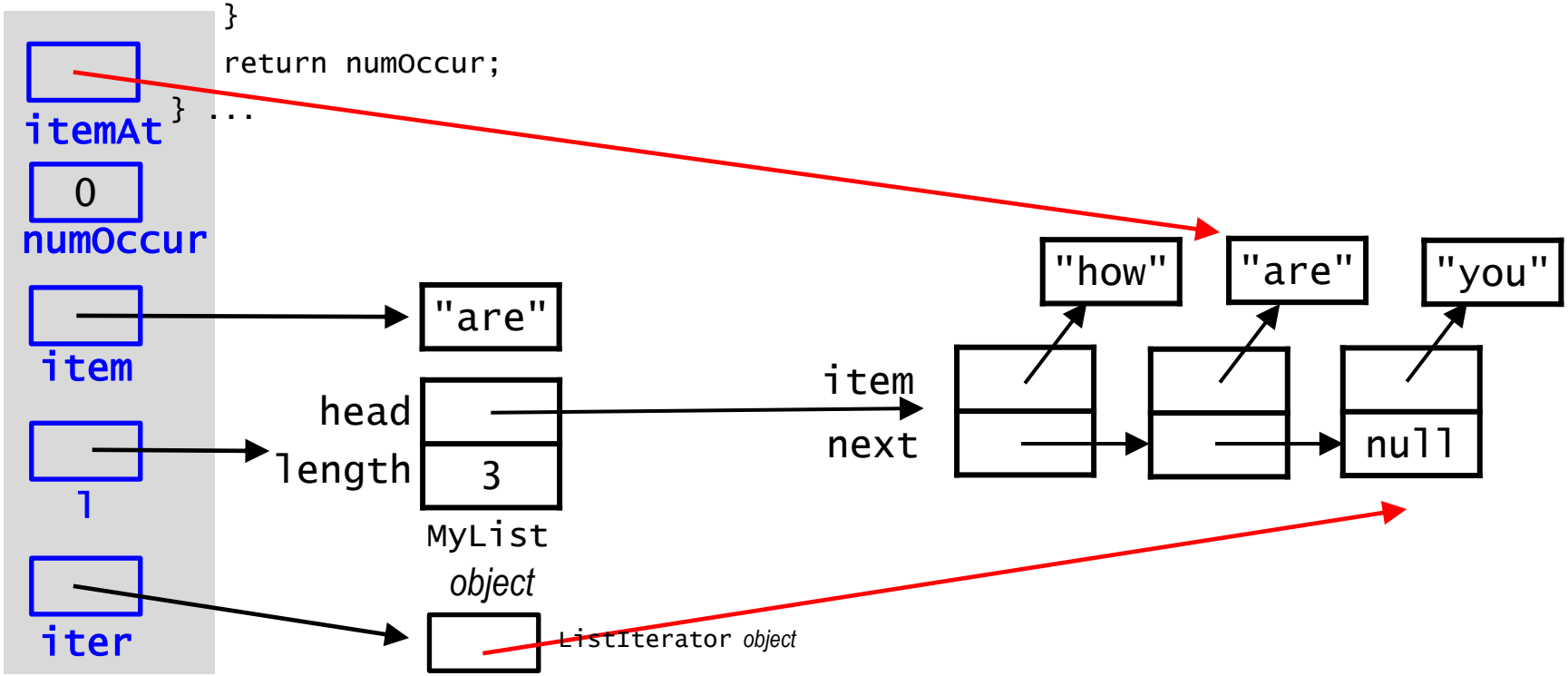

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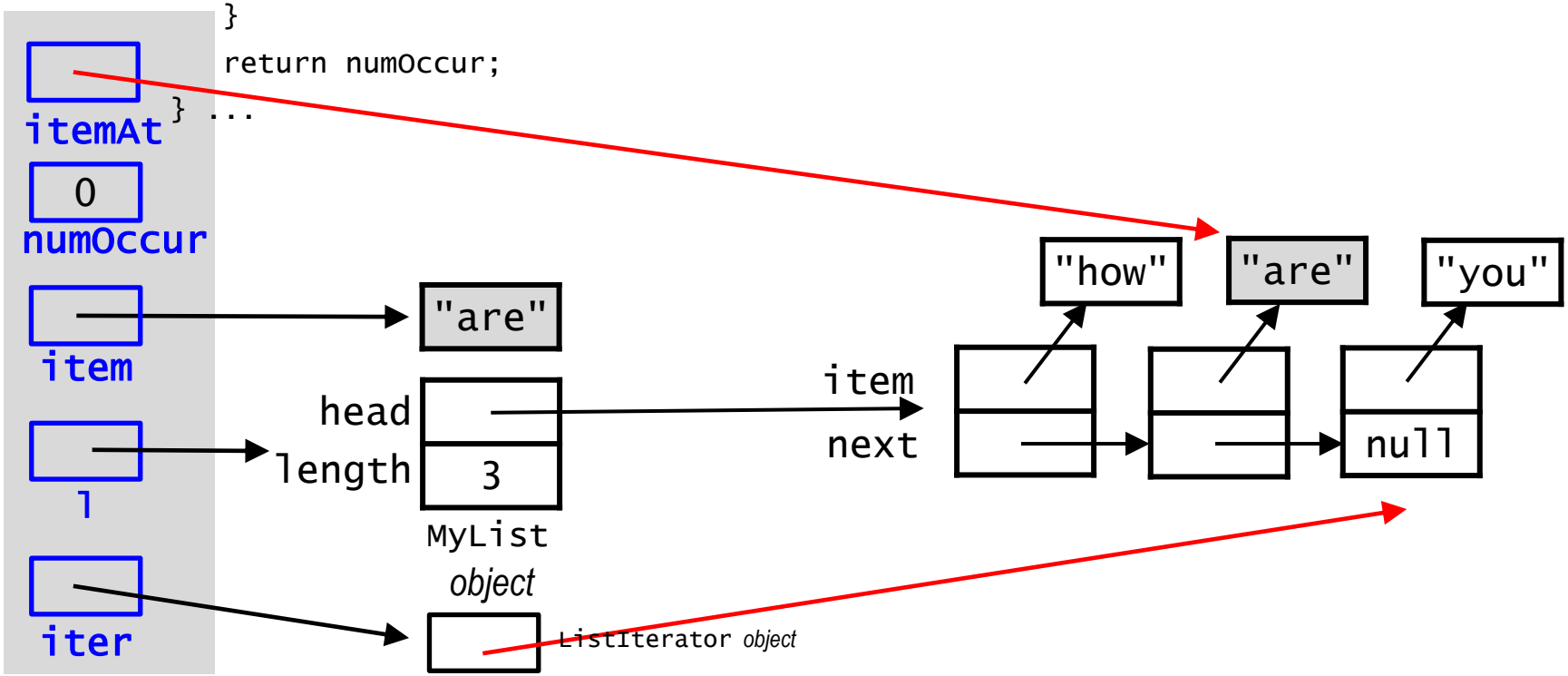

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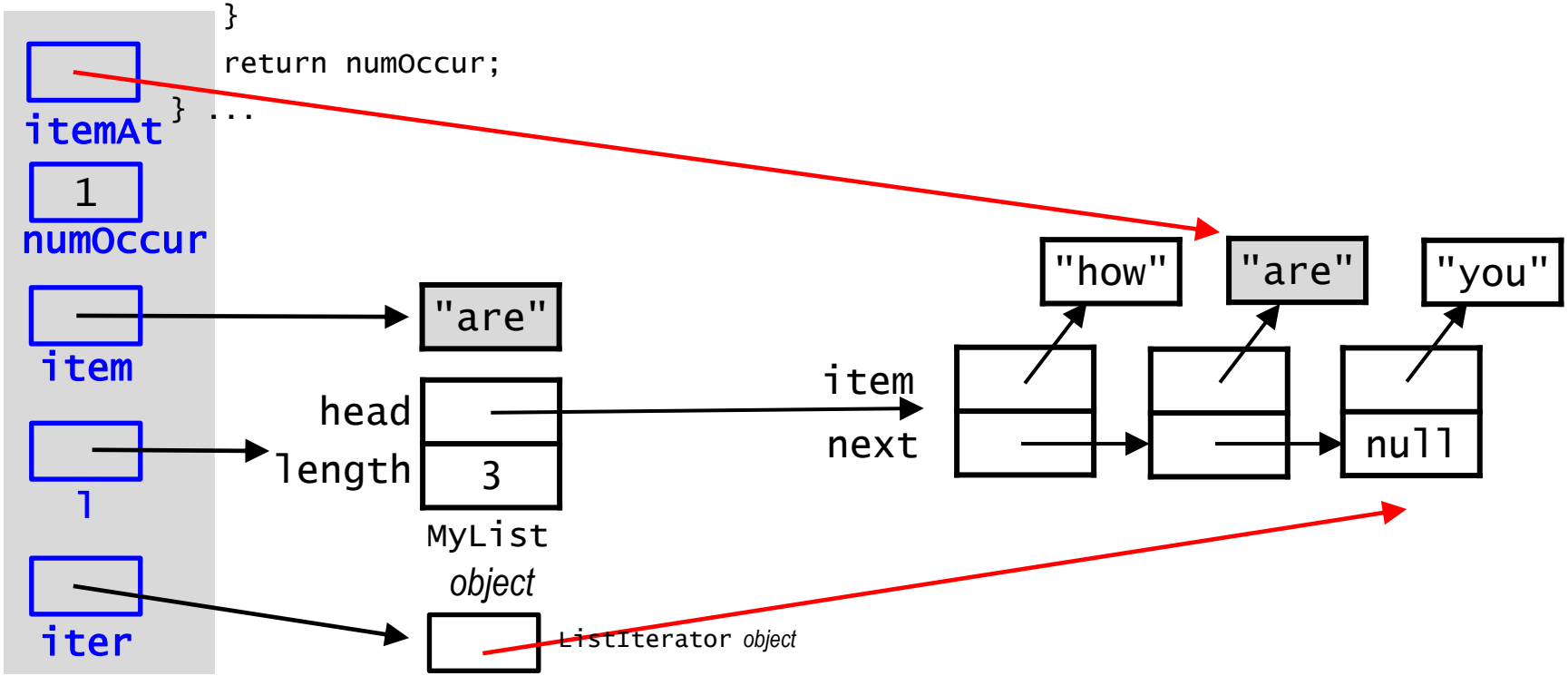

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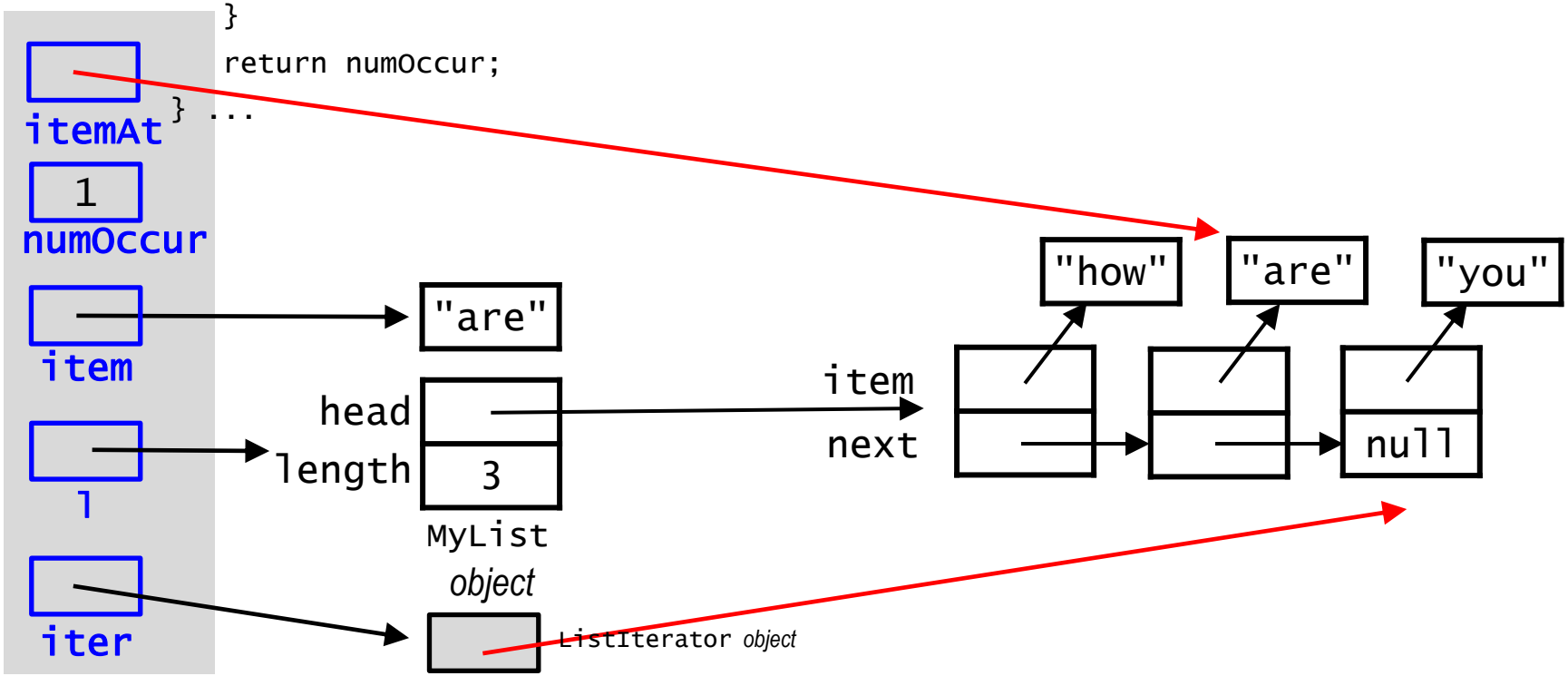

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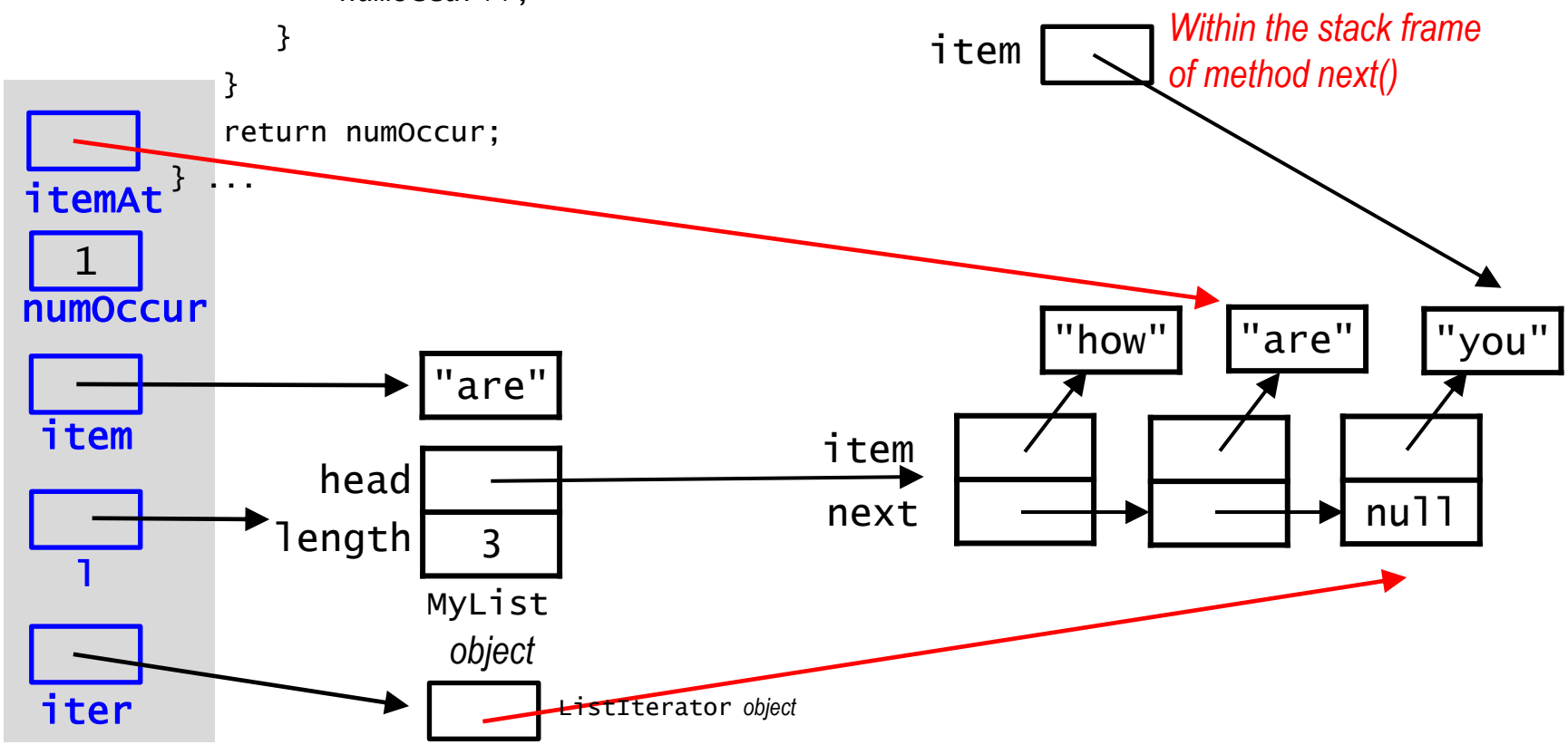

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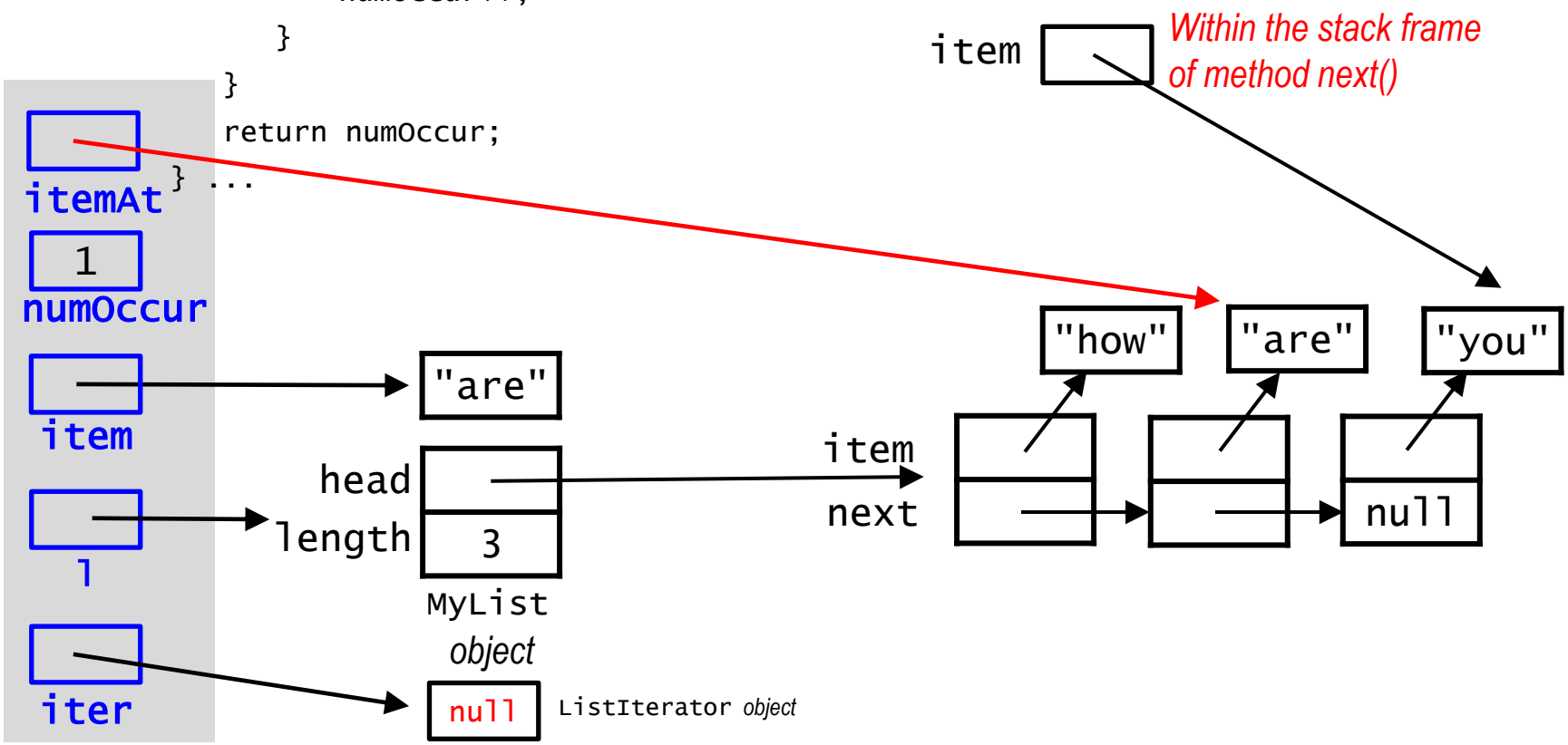

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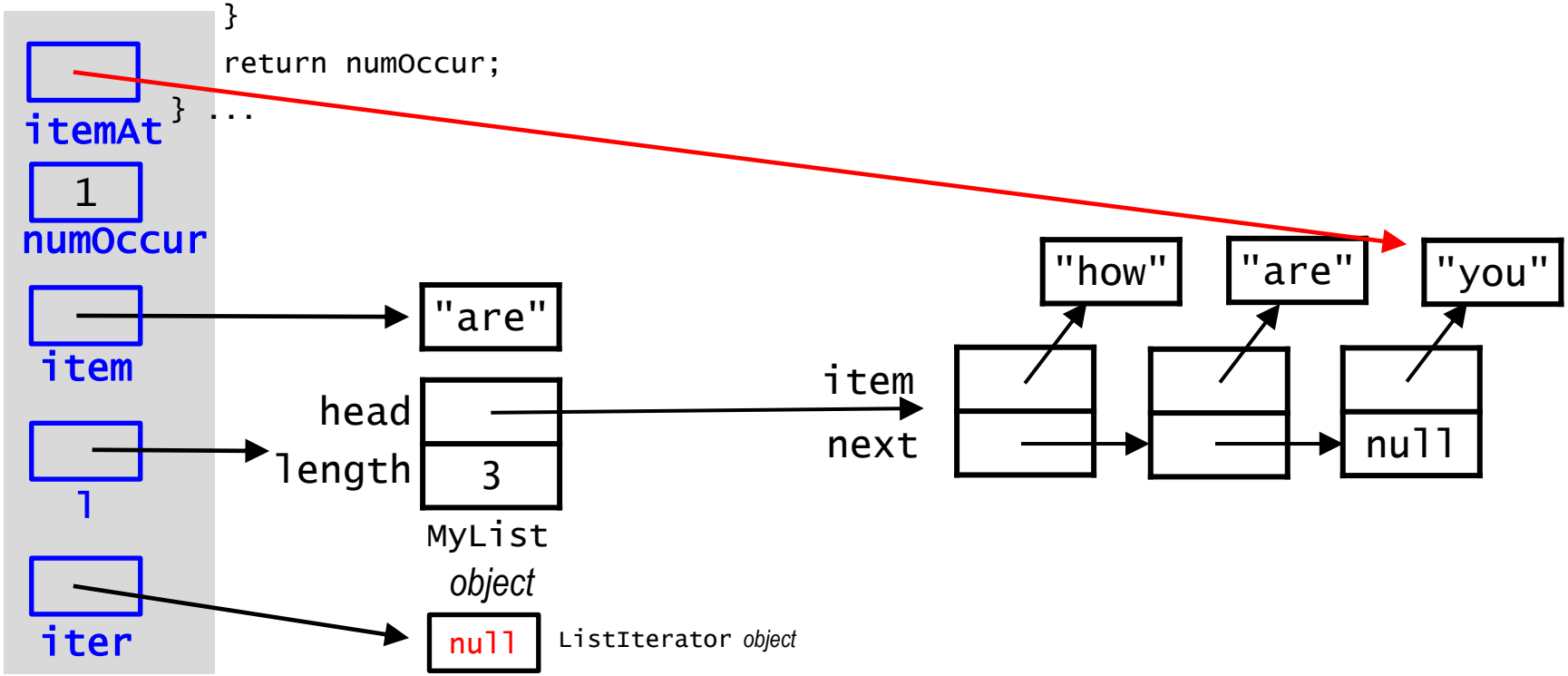

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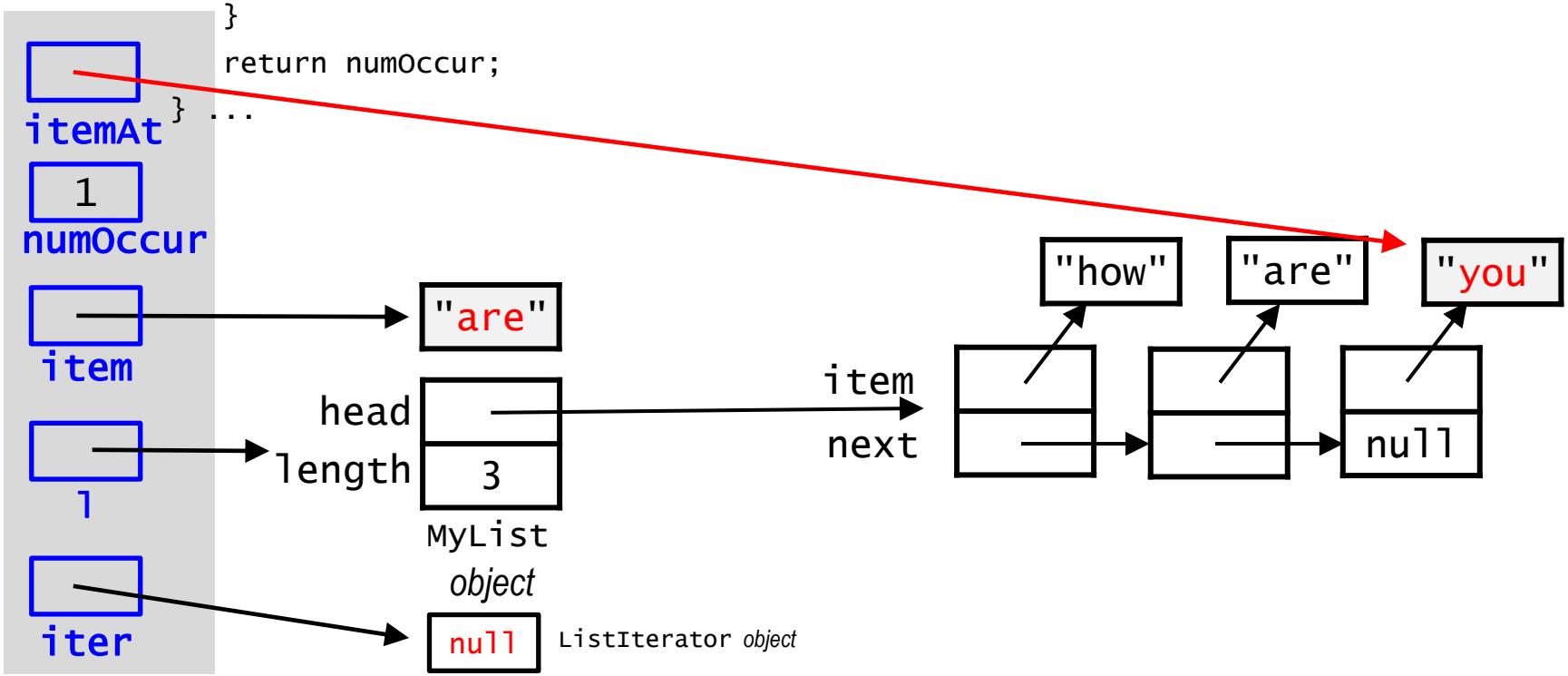

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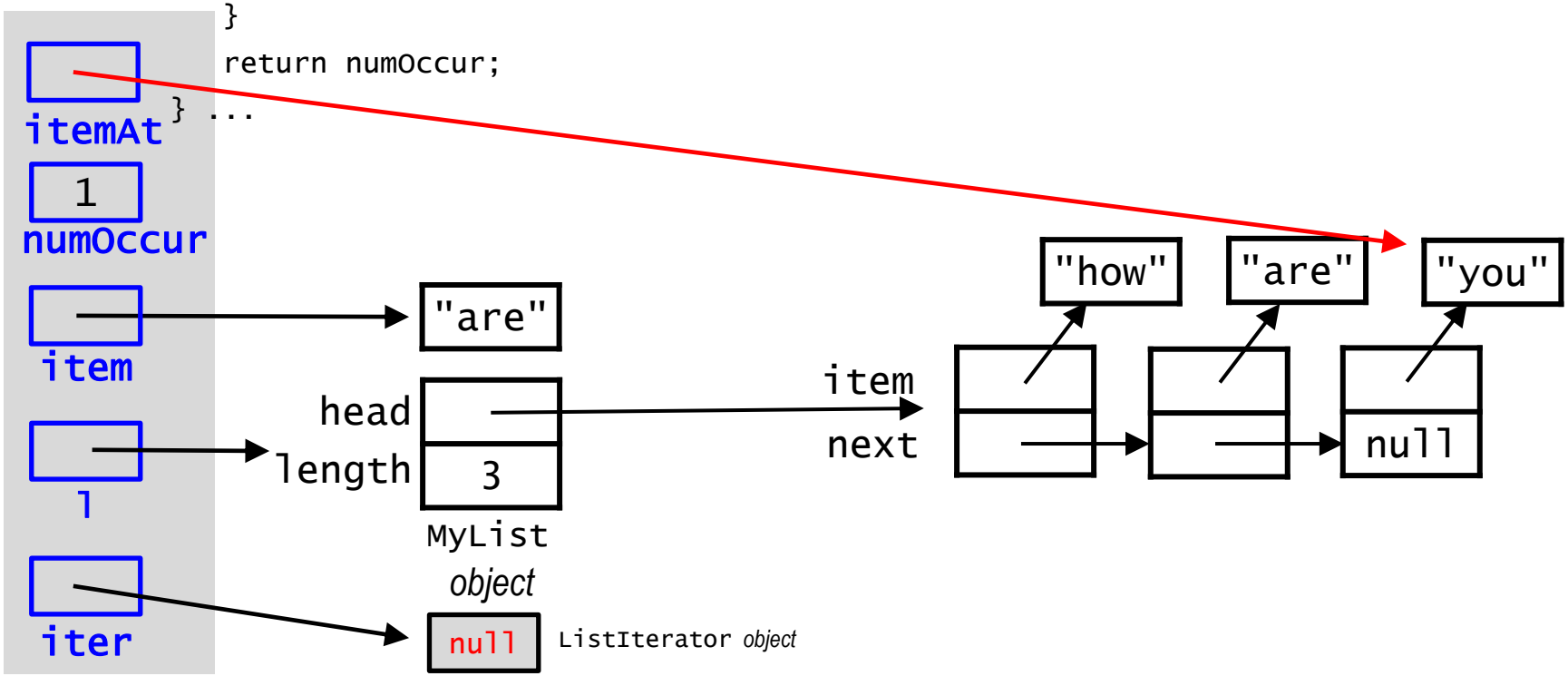

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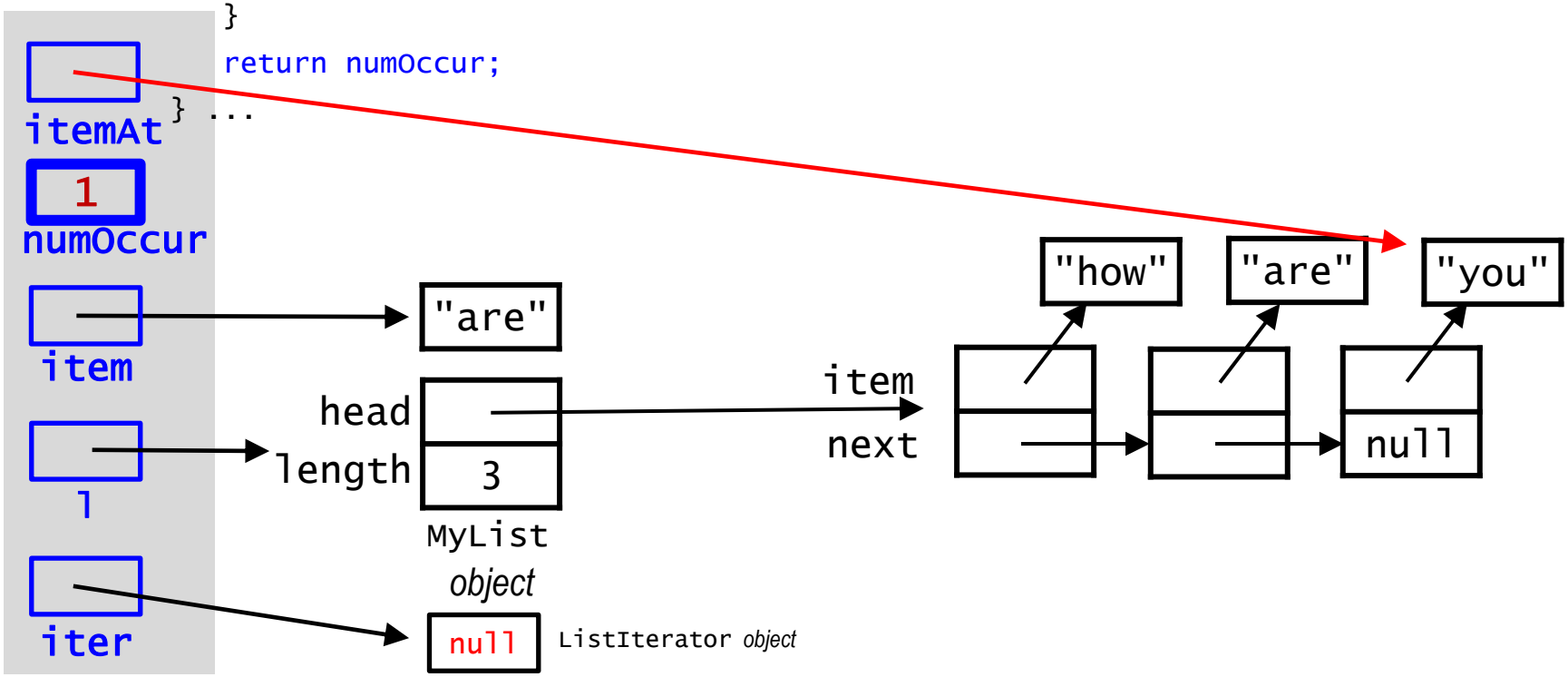

# An Interface for List Iterators:

## *summary*

Once the iterator interface has been implemented, we can create an instance of it and use it to externally traverse the list - regardless of the specific implementation of the List:

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         if (itemAt.equals(item)) {
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      }
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   } ...
```

*Note that this method can be called on any implementation of the List Interface.*
*What are the implications of this?*

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Once the iterator interface has been implemented, we can create an instance of it and use it to externally traverse the list - regardless of the specific implementation of the List:

```
public class MyClass {

   public static int numOccur(List l, Object item) {

      int numOccur = 0;
      ListIterator iter = l.iterator();
      while ( iter.hasNext() ) {
         Object itemAt = iter.next();

         if (itemAt.equals(item)) {
            numOccur++;
         }
      }
      return numOccur;
   } ...
```

*Note that this method can be called on any implementation of the List Interface.*

*What are the implications of this?*

# Collection of Students:

*example*

```
public class Students implements Iterable {
    private List<Student> students = null;

    public Students(){
        students = new ArrayList<Student>();
        students.add( new UndergraduateStudent() );
        students.add( new GraduateStudent() );
    }

    public Iterator iterator() {
        return students.iterator();
    }

    public static void main( String[] args ) {
        Students slist = new Students();
        Iterator iter = slist.iterator();

        while( iter.hasNext() )
            System.out.println( iter.next() );
    }
}
```

# Collection of Students:

*example*

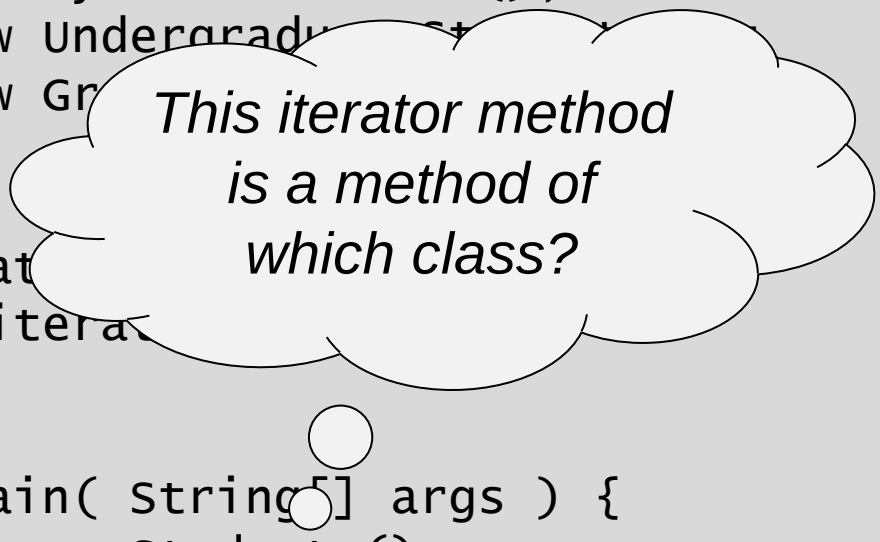
```
public class Students implements Iterable {
    private List<Student> students = null;

    public Students(){
        students = new ArrayList<Student>();
        students.add( new Undergrad
        students.add( new Gr
    }

    public Iterator iterat
        return students.itera
    }

    public static void main( String[] args ) {
        Students slist = new Students();
        Iterator iter = slist.iterator();

        while( iter.hasNext() )
            System.out.println( iter.next() );
    }
}
```



# Collection of Students:

*example*

```
public class Students implemen
    private List<Stude

    public Students()
        students = n
        students.add( new                    tudent() );
        students.add( new GraduateStudent() );
    }

    public Iterator iterator() {
        return students.iterator();
    }

    public static void main( String[] args ) {
        Students slist = new Students();
        Iterator iter = slist.iterator();

        while( iter.hasNext() )
            System.out.println( iter.next() );
    }
}
```

*This iterator method is a method of which class?*

# Collection of Students:

*example*

```
public class Students implements Iterable {
    private List<Student> students = null;

    public Students(){
        students = new ArrayList<Student>();
        students.add( new UndergraduateStudent() );
        students.add( new GraduateStudent() );
    }

    public Iterator iterator() {
        return students.iterator();
    }

    public static void main( String[] args ) {
        Students slist = new Students();
        Iterator iter = slist.iterator();

        while( iter.hasNext() )
            System.out.println( iter.next() );
    }
}
```

# Collection of Students:

*example*

```
public class Students implements Iterable {
    private List<Student> students = null;

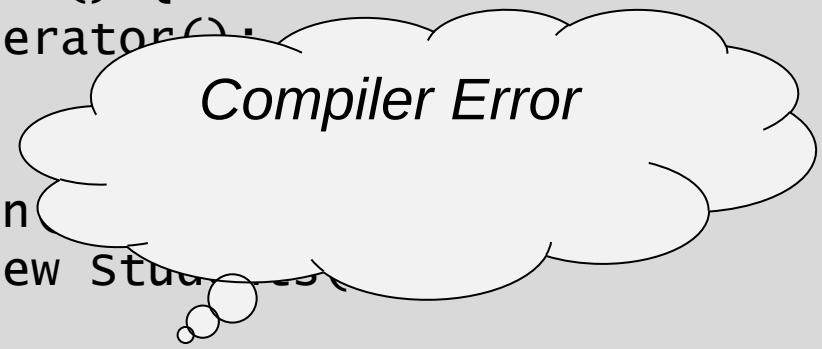
    public Students(){
        students = new ArrayList<Student>();;
        students.add( new UndergraduateStudent() );
        students.add( new GraduateStudent() );
    }

    public Iterator iterator() {
        return students.iterator();
    }

    public static void main
        Students slist = new Stu

        for (Student s : slist )
            System.out.println( s );

    }
}
```



# Collection of Students:

*example*

```
public class Students implements Iterable<Student> {
    private List<Student> students = null;

    public Students(){
        students = new ArrayList<Student>();;
        students.add( new UndergraduateStudent() );
        students.add( new GraduateStudent() );
    }

    public Iterator iterator() {
        return students.iterator();
    }

    public static void main( Str
        Students slist = new Stude

        for (Student s : slist )
            System.out.println( s );

    }
}
```

*Has to know that the method next() is returning a Student object.*

# Collection of Students:

*example*

```
public class Students implements Iterable<Student> {
    private List<Student> students = null;

    public Students(){
        students = new ArrayList<Student>();
        students.add( new UndergraduateStudent() );
        students.add( new GraduateStudent() );
    }

    public Iterator iterator() {
        return students.iterator();
    }

    public static void main
        Students slist =

        for (Student s :
            System.out.printl

    }
}
```

*Compiler warning of possible mistype.*

# Collection of Students:

*example*

```
public class Students implements Iterable<Student> {
    private List<Student> students = null;

    public Students(){
        students = new ArrayList<Student>();;
        students.add( new UndergraduateStudent() );
        students.add( new GraduateStudent() );
    }

    public Iterator<Student> iterator() {
        return students.iterator();
    }

    public static void main( String[] args ) {
        Students slist = new Students();

        for (Student s : slist )
            System.out.println( s );

    }
}
```

# Collection of Students:

*example*

```
public class Students implements Iterable<Student> {
    private List<Student> students = null;

    public Students(){
        students = new ArrayList<Student>();;
        students.add( new UndergraduateStudent() );
        students.add( new GraduateStudent() );
    }

    public Iterator<Student> iterator() {
        return students.iterator();
    }

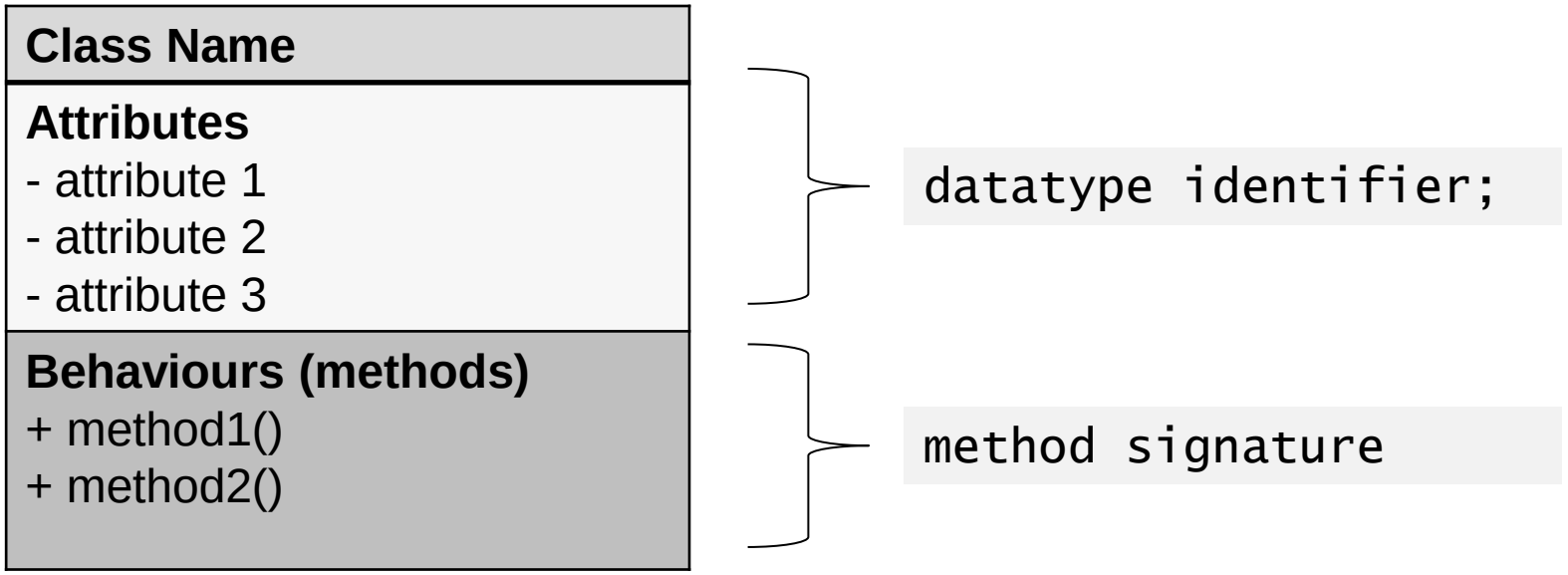
    public static void main( String[] args ) {
        Students slist = new Students();

        slist.forEach(System.out::println);


    }
}
```

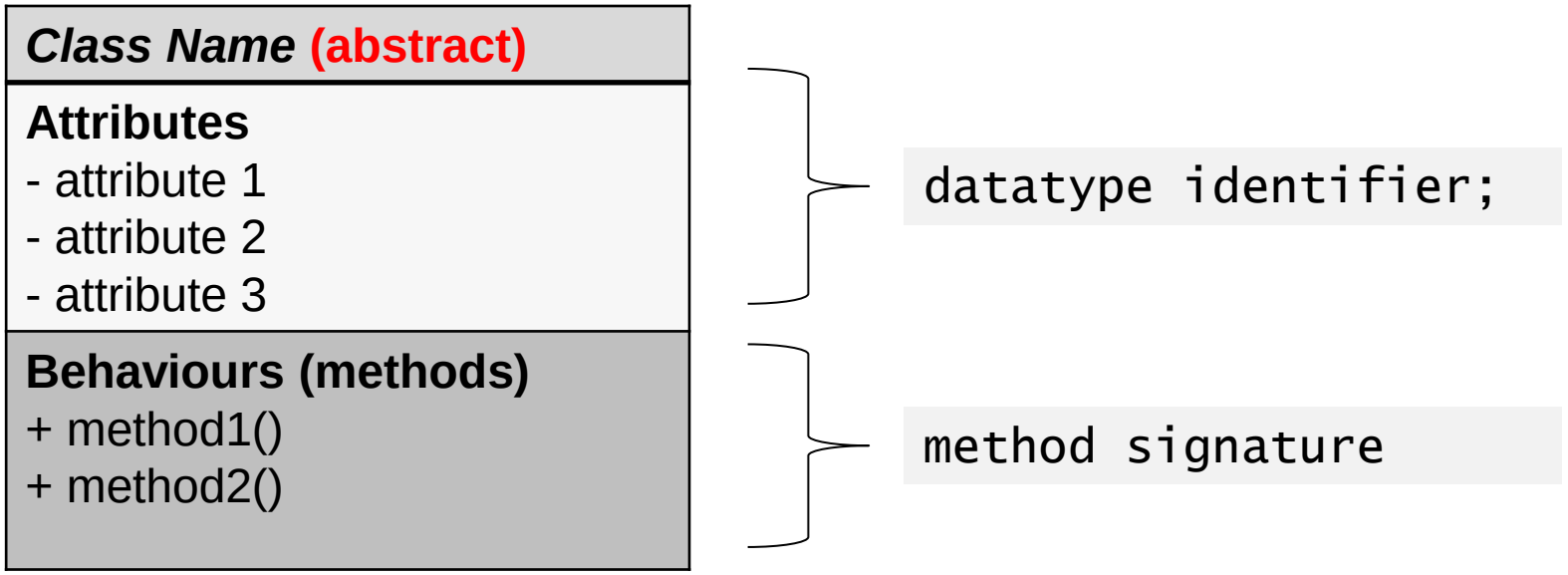
# UML Diagrams

UML Tutorial

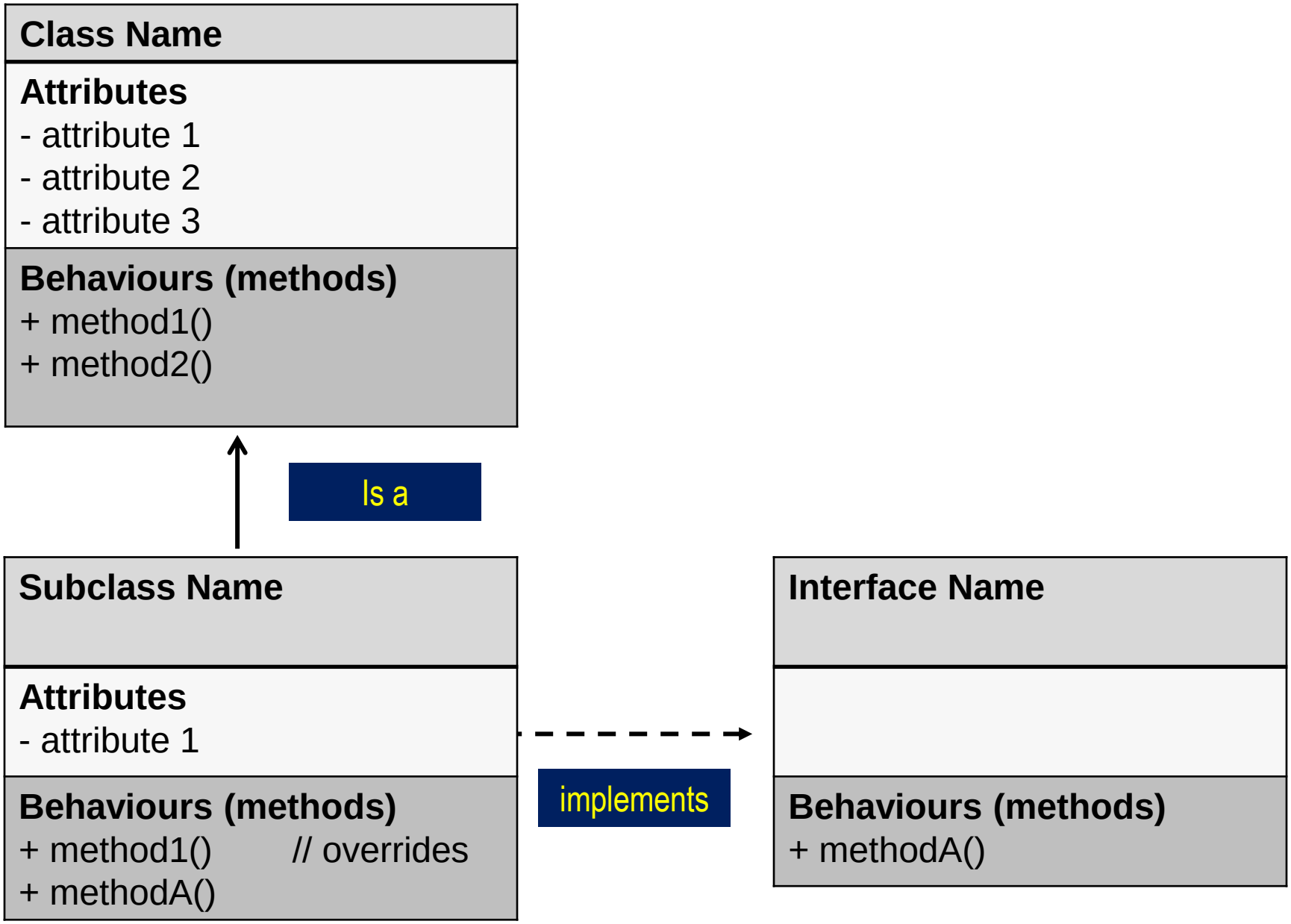


# UML Diagrams

UML Tutorial



# UML Diagrams



# Recall out Iterator Example